

Integrated SCFA and Microbiome Analysis: Abbott Carbohydrate Obesity Project

[Code ▾](#)

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1. Introduction

This integrated report brings together the short-chain fatty acid (SCFA) metabolomics and 16S rRNA gene microbiome analyses from the Lurie-Abbott carbohydrate obesity project. It re-runs the key components of each analysis within a single, reproducible R Markdown workflow and adds new integrative components relating taxa, SCFA concentrations, and responder phenotypes.

The goal is to maintain the rigor and depth of the original analyses while ensuring that metadata, filtering, and modeling decisions are harmonized across datasets.

2. Project Objectives

In this section, we summarize the main scientific and analytical objectives, building on the SCFA project objectives and extending them to explicitly include microbiome composition, diversity, and taxon-SCFA relationships.

3. Methods

This integrated analysis combines methods from both the SCFA metabolomics and 16S rRNA gene microbiome analyses. Below we describe the key methodological components employed throughout this report.

3.1 Study Design and Sample Processing

The study design includes two groups (Control/Healthy and Case/Obese) exposed to three carbohydrate conditions: no added carbohydrate (control), rapid digestible carbohydrate (RDC), and slow digestible carbohydrate (SDC). Samples were collected at multiple timepoints (0H, 48H, and 72H for SCFA; 0H and 48H for microbiome). All analyses account for repeated measures within subjects through appropriate statistical modeling.

3.2 SCFA Quantification and Preprocessing

Short-chain fatty acids (butyrate, propionate, acetate, and others) were quantified using gas chromatography-mass spectrometry (GC-MS) with derivatization. Raw concentration data were preprocessed to remove quality control samples, handle missing values, and standardize units. Concentrations are reported in micromolar (μM) units. Data were log-transformed where appropriate for statistical analyses.

3.3 Microbiome Sequencing, Processing, and Phyloseq Construction

16S rRNA gene sequencing was performed on fecal samples, and sequences were processed through standard quality control, denoising, and taxonomic assignment pipelines. The processed data were organized into a `phyloseq` object containing an OTU/ASV table, taxonomic classifications, sample metadata, and phylogenetic tree information. Relative abundances were calculated by normalizing counts to proportions within each sample. Absolute abundances were derived by integrating qPCR-determined total bacterial genome copies per sample with relative abundance data.

3.4 Statistical Modeling Strategies

3.4.1 Mixed-Effects Models for SCFA Concentrations

Linear mixed-effects models were fitted using `lmerTest::lmer()` to analyze SCFA concentrations as a function of group (Control vs Case), carbohydrate type, timepoint, and their interactions, with random intercepts for subject to account for repeated measures. Type III ANOVA tables were generated using `anova()` to test main effects and interactions, with p-values adjusted for multiple comparisons using the Benjamini-Hochberg (BH) false discovery rate (FDR) correction.

3.4.2 Alpha Diversity Analysis

Within-sample microbial diversity was assessed using three metrics: Observed richness (number of unique taxa), Shannon diversity index (accounts for both richness and evenness), and Simpson diversity index (emphasizes dominant taxa). Differences in alpha diversity between groups, carbohydrate conditions, and timepoints were tested using Wilcoxon rank-sum tests (Mann-Whitney U tests) with BH FDR correction for multiple comparisons.

3.4.3 Beta Diversity and Community Structure

Between-sample microbial community dissimilarity was quantified using Bray-Curtis distance on relative abundance data. Principal coordinates analysis (PCoA) was used to visualize community structure in reduced dimensions. Permutational multivariate analysis of variance (PERMANOVA) using `vegan::adonis2()` with 999 permutations was employed to test for significant differences in community composition between groups, carbohydrate conditions, and timepoints, with subject ID as a stratification variable to account for repeated measures.

3.4.4 Differential Abundance Analysis

Two complementary methods were used to identify taxa that differ significantly between conditions:

ANCOM-BC2: Analysis of Compositions of Microbiomes with Bias Correction 2 was applied using the `ANCOMBC` package. This method accounts for compositionality and provides log-fold change estimates with bias correction, standard errors, and FDR-adjusted q-values. Models included group, carbohydrate, timepoint, and their interactions as fixed effects, with subject ID as a random effect where appropriate.

MaAsLin2: Microbiome Multivariable Association with Linear Models 2 was applied using the `Maaslin2` package. This method uses linear models with total sum scaling (TSS) normalization and log transformation. It provides coefficient estimates, standard errors, p-values, and FDR-adjusted q-values. Models included the same fixed and random effects as ANCOM-BC2 analyses.

Both methods were applied to relative abundance data, and results were filtered to retain associations with $q < 0.1$ (FDR-adjusted p-value threshold).

3.4.5 Taxon–SCFA Association Analysis

To identify microbial taxa associated with SCFA concentrations, ANCOM-BC2 and MaAsLin2 were applied with butyrate or propionate concentrations as continuous predictors, along with carbohydrate type and their interactions. These analyses were performed on relative abundance data and included subject ID as a random effect to account for repeated measures.

3.4.6 Correlation Analyses

Spearman rank correlation coefficients were calculated to assess monotonic relationships between taxon abundances (e.g., *Fusicatenibacter*) and SCFA concentrations. Linear regression models were fitted to estimate R^2 values and test for linear associations. Correlations were examined at specific timepoints (e.g., 48H) and carbohydrate conditions (e.g., SDC only) to focus on biologically relevant contexts.

3.4.7 Responder vs Non-Responder Classification

Responder status was defined separately for butyrate and propionate based on the change in SCFA concentration from baseline (0H) to 48H ($\Delta = 48H - 0H$). For each analyte and carbohydrate condition, samples were classified as “Responder” if their delta value was greater than or equal to the median delta, and “Non-Responder” if below the median. This data-driven threshold allows for identification of individuals who

show a substantial SCFA response to carbohydrate challenge relative to their peers in the same condition.

3.4.8 Responder Group Comparisons

Microbial composition differences between responder and non-responder groups were assessed using: - Alpha diversity comparisons with Wilcoxon rank-sum tests - Beta diversity comparisons using PERMANOVA - Differential abundance analysis using ANCOM-BC2 and MaAsLin2 to identify taxa that differ between responder and non-responder groups

3.5 Integration Methods

SCFA and microbiome data were integrated by matching samples based on subject ID, timepoint, and carbohydrate condition. Metadata were harmonized to ensure consistent group labels (Control/Healthy, Case/Obese) and carbohydrate labels (No Added Carb, RDC, SDC) across datasets. The integrated metadata were added to the phyloseq object's sample data component to enable joint analyses.

3.6 Visualization and Reporting

All visualizations were created using `ggplot2` with a consistent color palette and theme derived from the SCFA analysis. Forest plots display effect sizes (log-fold changes) with 95% confidence intervals for ANCOM-BC2 results. Volcano plots show effect sizes versus significance for MaAsLin2 results. Alluvial diagrams illustrate compositional changes over time. Statistical test results are displayed directly on plots where appropriate (e.g., Wilcoxon p-values on boxplots, R^2 values on scatter plots). All p-values are reported with FDR adjustment where multiple comparisons were performed.

4. Summary Data

We begin with a concise overview of the sample counts by obesity status (Control vs Case) and carbohydrate condition across both SCFA and microbiome datasets.

SCFA samples by group and carbohydrate

Group	Carbohydrate	N samples
Control	No Added Carb	16
Control	RDC	16
Control	SDC	16

Group	Carbohydrate	N samples
Case	No Added Carb	16
Case	RDC	16
Case	SDC	16

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Microbiome samples by group, carbohydrate, and timepoint

Group	Carbohydrate	Timepoint	N samples
Healthy	No Added Carb	0H	12
Healthy	No Added Carb	48H	14
Healthy	RDC	0H	28
Healthy	RDC	48H	27
Healthy	SDC	0H	27
Healthy	SDC	48H	28
Obese	No Added Carb	0H	13
Obese	No Added Carb	48H	19
Obese	RDC	0H	28
Obese	RDC	48H	35

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5. Metabolomics (SCFA)

5.1 Section Overview

This section reproduces and extends the SCFA analyses, focusing on:

- Carbohydrate type effects (combined and stratified by group)
- SCFA concentrations over time
- Advanced mixed-effects modeling for SCFA trajectories

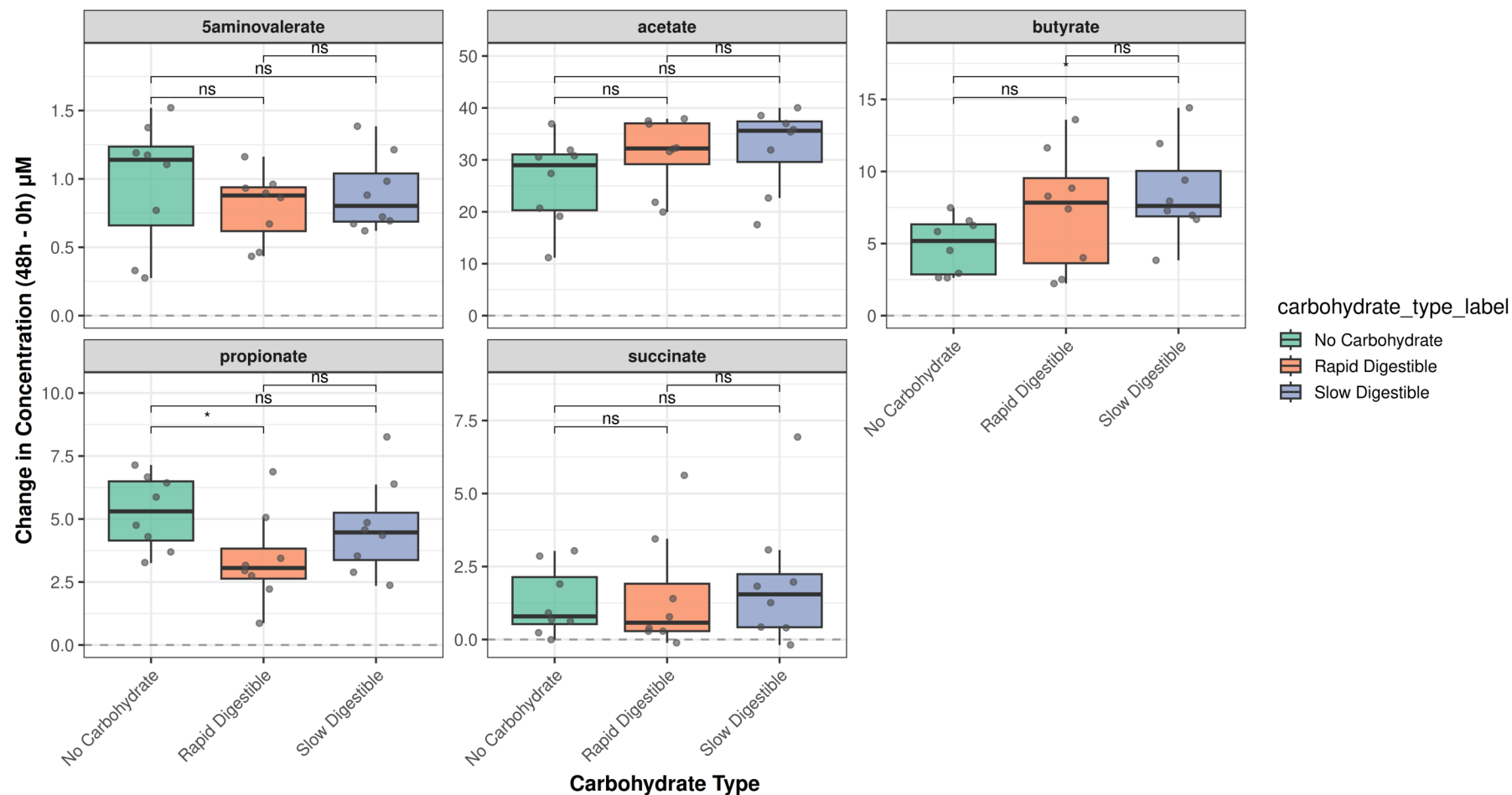
We first quantify SCFA response magnitudes (48h – 0h) and evaluate how they vary by carbohydrate type and obesity group. We then visualize SCFA concentrations over time and fit linear mixed-effects models with subject as a random effect to formally test fixed effects of group, carbohydrate type, time, and their interactions on absolute SCFA concentrations.

5.2 Carbohydrate Type Effects (Combined Groups)

5.2.1 Control Group: Delta SCFA by Carbohydrate Type

SCFA Response Magnitude by Carbohydrate Type - Control Group Only

Delta change (48h - 0h) with Wilcoxon pairwise comparisons



SCFA Response Magnitude by Carbohydrate Type - Case Group Only

5aminovalerate

ns

ns

ns

acetate

ns

ns

butyrate

ns

ns

propionate

ns

*

succinate

ns

ns

ns

Change in Concentration (48h - 0h) μ M

No Carbohydrate Rapid Digestible Slow Digestible

Carbohydrate Type

SCFA Response Magnitude by Carbohydrate Type (Combined Groups)

Figure 2 displays the change in concentration of short-chain fatty acids (SCFAs) after 48h of incubation, comparing three carbohydrate types: No Carbohydrate, Rapid Digestible, and Slow Digestible. The y-axis represents the Change in Concentration (48h - 0h) μM . The x-axis is labeled Carbohydrate Type. The legend indicates the carbohydrate type labels: No Carbohydrate (green), Rapid Digestible (orange), and Slow Digestible (blue).

The figure consists of five box plots arranged in a 2x3 grid. The top row shows 5aminovaleate, acetate, and butyrate. The bottom row shows propionate and succinate. Each plot compares three carbohydrate types: No Carbohydrate (green), Rapid Digestible (orange), and Slow Digestible (blue). The y-axis is 'Change in Concentration (48h - 0h) μM '. Statistical significance is indicated by brackets and labels: 'ns' for not significant and '**' for $p < 0.01$. A dashed line at y=0 indicates no change.

SCFA	Carbohydrate Type	Median Change (μM)	Significance (vs. No Carbohydrate)
5aminovaleate	No Carbohydrate	~0.8	-
	Rapid Digestible	~0.7	ns
	Slow Digestible	~0.7	ns
acetate	No Carbohydrate	~28	-
	Rapid Digestible	~32	ns
	Slow Digestible	~35	ns
butyrate	No Carbohydrate	~5.5	-
	Rapid Digestible	~7.5	**
	Slow Digestible	~7.5	ns
propionate	No Carbohydrate	~5.5	-
	Rapid Digestible	~3.0	**
	Slow Digestible	~4.0	ns
succinate	No Carbohydrate	~0.5	-
	Rapid Digestible	~1.0	ns
	Slow Digestible	~1.5	ns

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analyste	Effect	p	p_adj	significance
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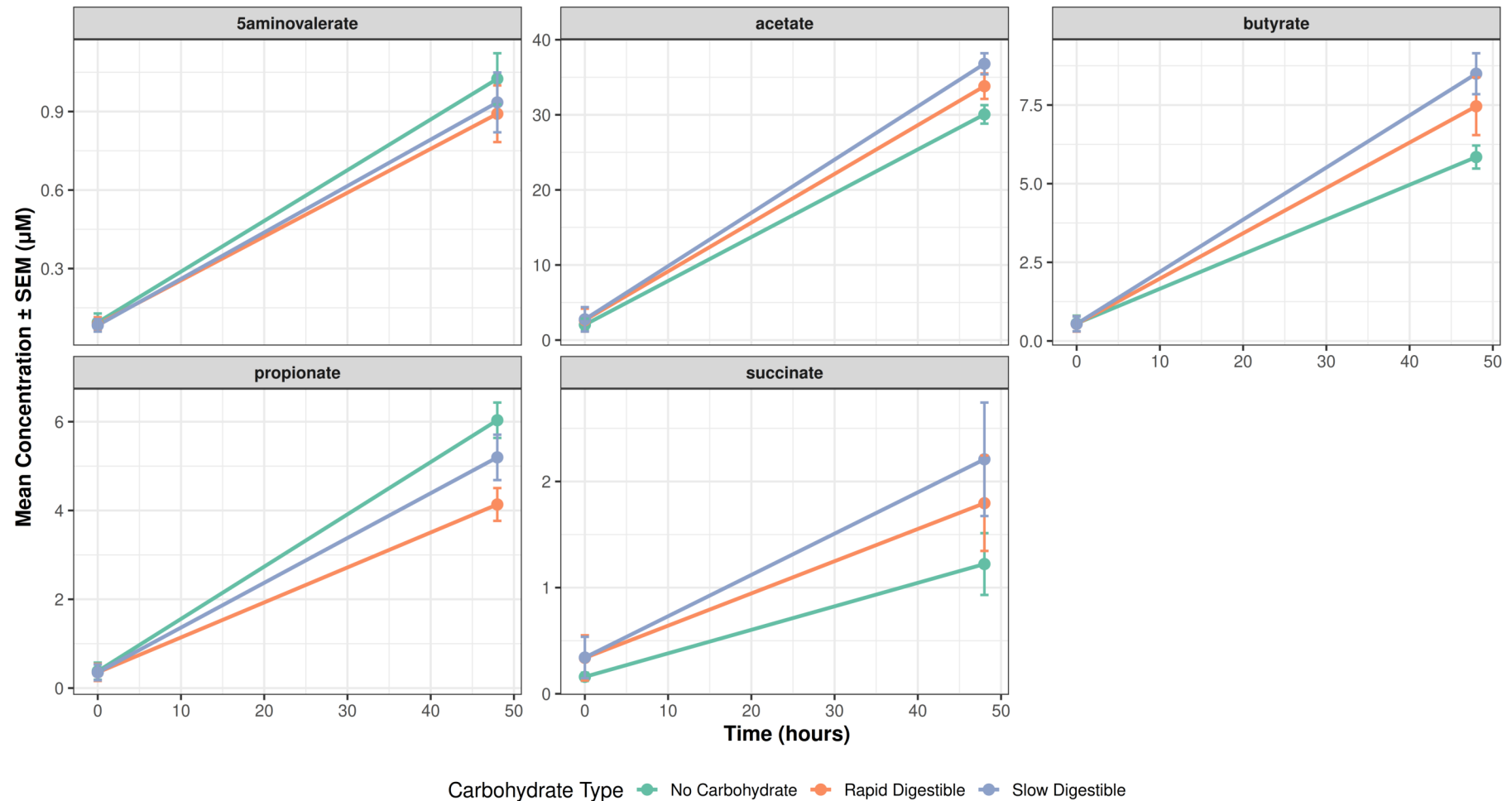
5aminovalelate	carbohydrate_type	0.7032	0.7032	ns
5aminovalelate	Residuals			ns
acetate	carbohydrate_type	0.0479	0.0479	p<0.05
acetate	Residuals			ns
butyrate	carbohydrate_type	0.0420	0.0420	p<0.05
butyrate	Residuals			ns
propionate	carbohydrate_type	0.0209	0.0209	p<0.05
propionate	Residuals			ns
succinate	carbohydrate_type	0.3920	0.3920	ns
succinate	Residuals			ns

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5.3 SCFA Concentrations Over Time

SCFA Concentrations Over Time by Carbohydrate Type



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5.4 Advanced Statistical Modeling: Mixed-Effects Models

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Search:

Mixed-Effects Models for SCFA Concentrations: ANOVA Table

analyte	effect	F_value	pr_f	p_adj	significance
acetate	timepoint_hr	1,294.3303	0.0000	0.0000	p<0.001
propionate	timepoint_hr	472.2423	0.0000	0.0000	p<0.001
butyrate	timepoint_hr	326.8937	0.0000	0.0000	p<0.001
5aminovalerate	timepoint_hr	259.2812	0.0000	0.0000	p<0.001
succinate	timepoint_hr	46.8123	0.0000	0.0000	p<0.001
propionate	carbohydrate_type	6.4607	0.0027	0.0084	p<0.01
propionate	carbohydrate_type:timepoint_hr	6.1058	0.0036	0.0084	p<0.01
acetate	carbohydrate_type	6.2720	0.0031	0.0109	p<0.05
butyrate	carbohydrate_type	4.2545	0.0180	0.0421	p<0.05
butyrate	carbohydrate_type:timepoint_hr	4.3484	0.0166	0.0421	p<0.05

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5.5 Summary of SCFA Results

The SCFA metabolomics analysis reveals several key findings regarding short-chain fatty acid production in response to carbohydrate challenge in adolescents with and without obesity:

1. Dominant Effect of Carbohydrate Type and Time

The primary drivers of SCFA production in this ex vivo model were the type of carbohydrate supplied and the incubation time. Both rapid digestible carbohydrate (RDC) and slow digestible carbohydrate (SDC) led to significant increases in all measured SCFAs (butyrate, propionate, acetate) over 48 hours compared to the no-carbohydrate control. This was confirmed by: - Significant differences in delta changes (48H - 0H) between carbohydrate conditions across multiple SCFAs (Section 5.2) - Clear temporal increases in SCFA concentrations over time, with distinct trajectories for each carbohydrate type (Section 5.3) - Robust statistical significance in the mixed-effects models, where carbohydrate type and timepoint consistently emerged as significant main effects and interactions (Section 5.4)

2. Subtle Differences Between Case and Control Groups

While there were some minor differences in baseline SCFA levels and response patterns between the case (obesity) and control groups, the overall response to carbohydrate interventions was remarkably similar. The mixed-effects models showed limited significant main effects of group or major interactions involving group, suggesting that the fundamental capacity of the gut microbiota to ferment these carbohydrates is not dramatically altered in the context of obesity in this cohort. This finding suggests that obesity status alone does not substantially impair the microbial fermentation capacity for these carbohydrate substrates.

3. Carbohydrate-Specific Response Patterns

The analyses revealed distinct response patterns for different carbohydrate types: - Both RDC and SDC stimulated SCFA production compared to no-carbohydrate controls - The magnitude and timing of responses varied by carbohydrate type, as evidenced by significant carbohydrate × timepoint interactions in the mixed-effects models - These differences highlight the importance of carbohydrate structure and digestibility in shaping microbial metabolic output

4. Pronounced Interpersonal Variation

A central finding is the high degree of subject-to-subject variability in SCFA production. The delta change analyses (Section 5.2) and temporal trajectories (Section 5.3) clearly demonstrate that: - Some individuals are consistently high or low SCFA producers across conditions - Response patterns to different carbohydrates are unique to each individual - This variability is evident even when accounting for group (case vs control) and carbohydrate type effects

This pronounced interpersonal variation underscores the need for personalized approaches in nutrition, as a “one-size-fits-all” dietary intervention is unlikely to be effective. The identification of responder and non-responder groups (explored in Section 7) builds upon this finding to enable more targeted analyses of microbial composition differences.

5. Statistical Model Insights

The linear mixed-effects models (Section 5.4) provide formal statistical confirmation of these patterns, accounting for repeated measures within subjects. Key model findings include: - Significant main effects of carbohydrate type and timepoint across multiple SCFAs - Significant interactions between carbohydrate type and timepoint, indicating that the temporal response depends on the carbohydrate substrate - Limited group effects, consistent with the observation that obesity status does not dramatically alter fermentation capacity - The models account for subject-level variation through random intercepts, capturing the interpersonal differences observed in the data

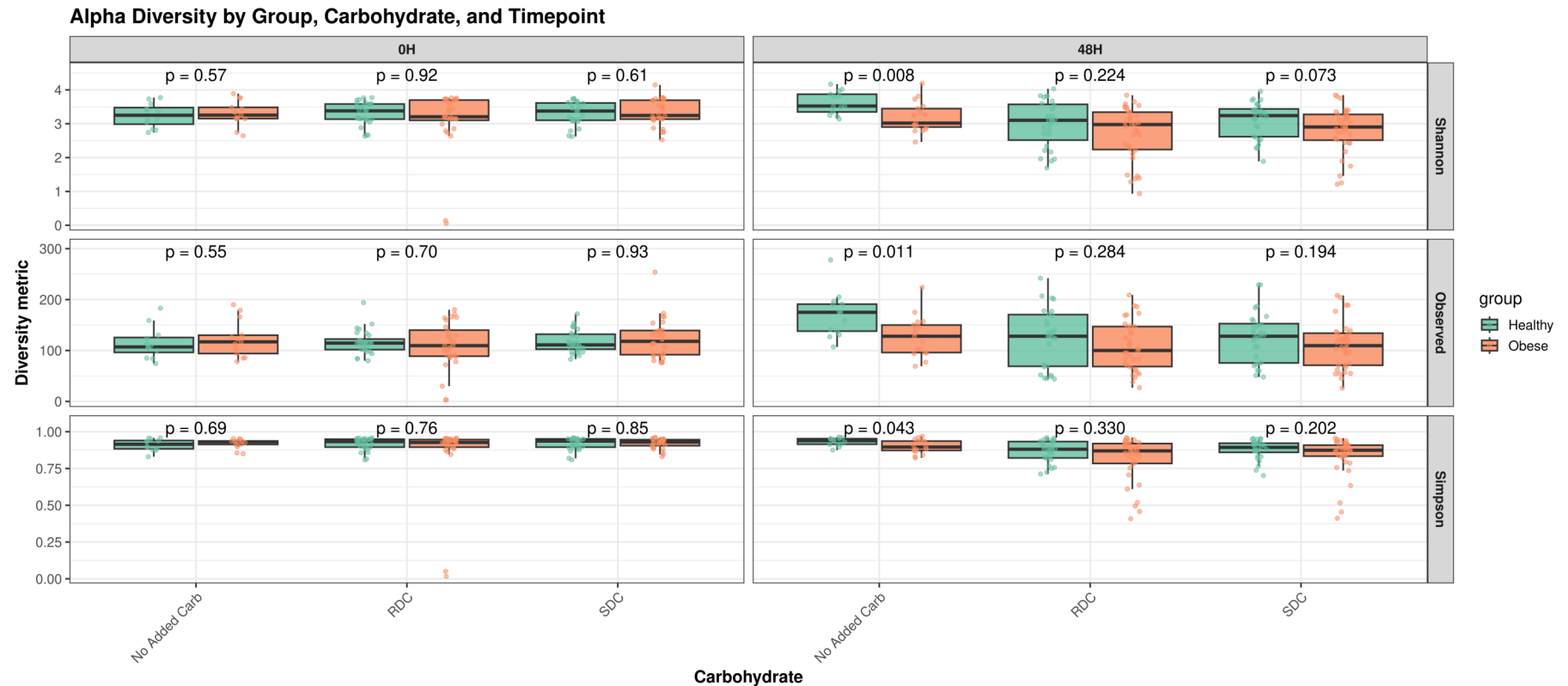
These findings set the stage for the integrated analyses in Section 7, where we examine how microbial composition relates to SCFA production patterns and identify specific taxa associated with SCFA concentrations and responder phenotypes.

6. Microbiome

6.1 Section Overview

This section reproduces key microbiome results, including alpha and beta diversity, community composition, and differential abundance, using the harmonized metadata and styling. We focus first on within-sample diversity (alpha diversity) to summarize richness and evenness patterns across groups, carbohydrate conditions, and timepoints, then move to between-sample diversity (beta diversity) and taxon-level changes.

6.2 Alpha Diversity



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Search:

Alpha diversity at 48H: summary by metric, group, and carbohydrate

metric	group	carbohydrate	mean	sd	n
Shannon	Healthy	No Added Carb	3.626	0.337	13

Shannon	Healthy	RDC	2.989	0.685	27
Shannon	Healthy	SDC	3.099	0.528	28
Shannon	Obese	No Added Carb	3.191	0.444	17
Shannon	Obese	RDC	2.712	0.793	35
Shannon	Obese	SDC	2.815	0.665	36
Observed	Healthy	No Added Carb	170.462	44.612	13
Observed	Healthy	RDC	125.333	58.541	27
Observed	Healthy	SDC	124.321	50.030	28
Observed	Obese	No Added Carb	127.588	39.391	17

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Search:

Wilcoxon tests: Healthy vs Obese within carbohydrate × timepoint for each alpha metric

metric	carbohydrate	timepoint	comparison	p value	p adjusted	significance
Shannon	No Added Carb	0H	Healthy vs Obese	0.5740	0.9000	ns
Shannon	No Added Carb	48H	Healthy vs Obese	0.0080	0.1017	ns
Shannon	RDC	0H	Healthy vs Obese	0.9160	0.9330	ns
Shannon	RDC	48H	Healthy vs Obese	0.2240	0.5760	ns
Shannon	SDC	0H	Healthy vs Obese	0.6100	0.9000	ns
Shannon	SDC	48H	Healthy vs Obese	0.0728	0.3276	ns

Observed	No Added Carb	0H	Healthy vs Obese	0.5490	0.9000	ns
Observed	No Added Carb	48H	Healthy vs Obese	0.0113	0.1017	ns
Observed	RDC	0H	Healthy vs Obese	0.7000	0.9000	ns
Observed	RDC	48H	Healthy vs Obese	0.2840	0.6390	ns

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```
##
##
## ##### Wilcoxon tests: Carbohydrate comparisons within group x timepoint
```

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Search:

Wilcoxon tests: Carbohydrate comparisons within group x timepoint for each alpha metric

metric	group	timepoint	comparison	p value	p adjusted	significance
Shannon	Healthy	0H	No Added Carb vs RDC	0.5500	0.8250	ns
Shannon	Healthy	0H	No Added Carb vs SDC	0.5380	0.8250	ns
Shannon	Healthy	0H	RDC vs SDC	0.9670	0.9670	ns
Shannon	Healthy	48H	No Added Carb vs RDC	0.0020	0.0060	**
Shannon	Healthy	48H	No Added Carb vs SDC	0.0040	0.0060	**
Shannon	Healthy	48H	RDC vs SDC	0.5990	0.5990	ns
Shannon	Obese	0H	No Added Carb vs RDC	0.6280	0.9670	ns
Shannon	Obese	0H	No Added Carb vs SDC	0.9670	0.9670	ns

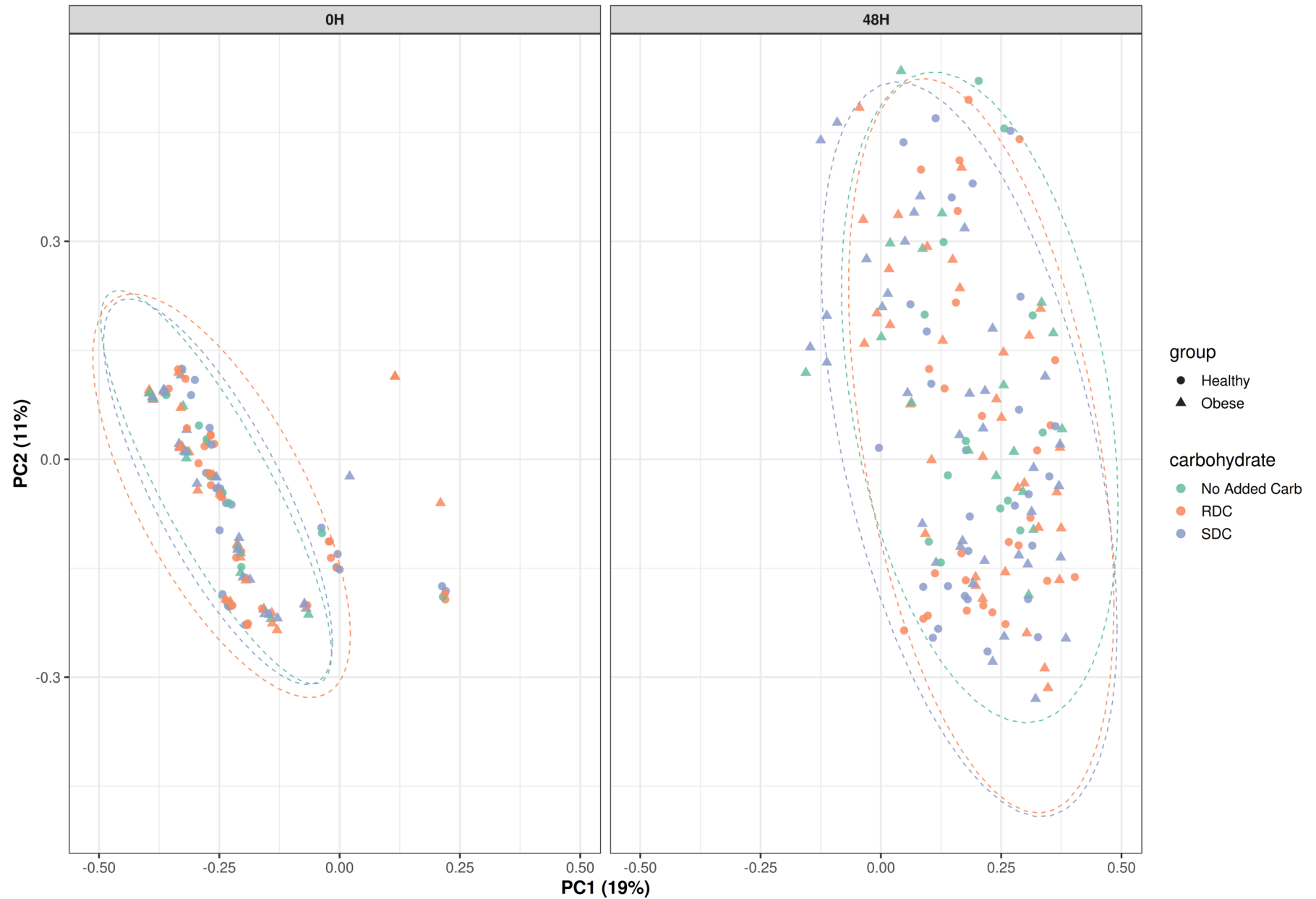
Shannon	Obese	0H	RDC vs SDC	0.6780	0.9670	ns
Shannon	Obese	48H	No Added Carb vs RDC	0.0980	0.1460	ns

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6.3 Beta Diversity

PCoA – Bray-Curtis distance



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Search:

PERMANOVA results (Bray-Curtis; all timepoints, subject-level strata)

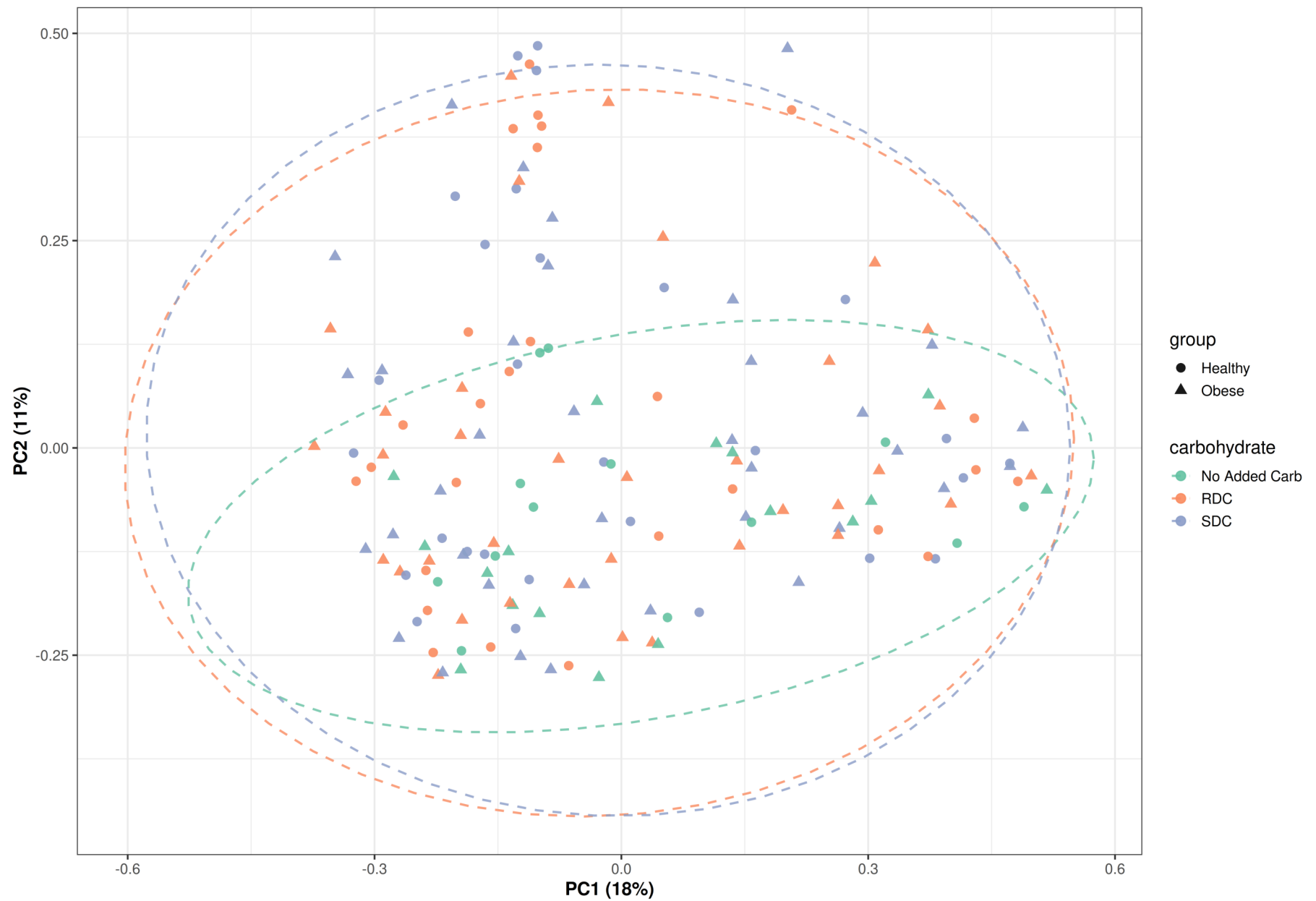
term	f	r2	p_value
Model	9.829	0.171	0.001
Residual		0.829	
Total		1.000	

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PCoA – Bray-Curtis distance (48H samples)

Show 10 entries

Search:

PERMANOVA results (Bray-Curtis; 48H only, subject-level strata)

term	f	r2	p_value
Model	1.470	0.047	0.001
Residual		0.953	
Total		1.000	

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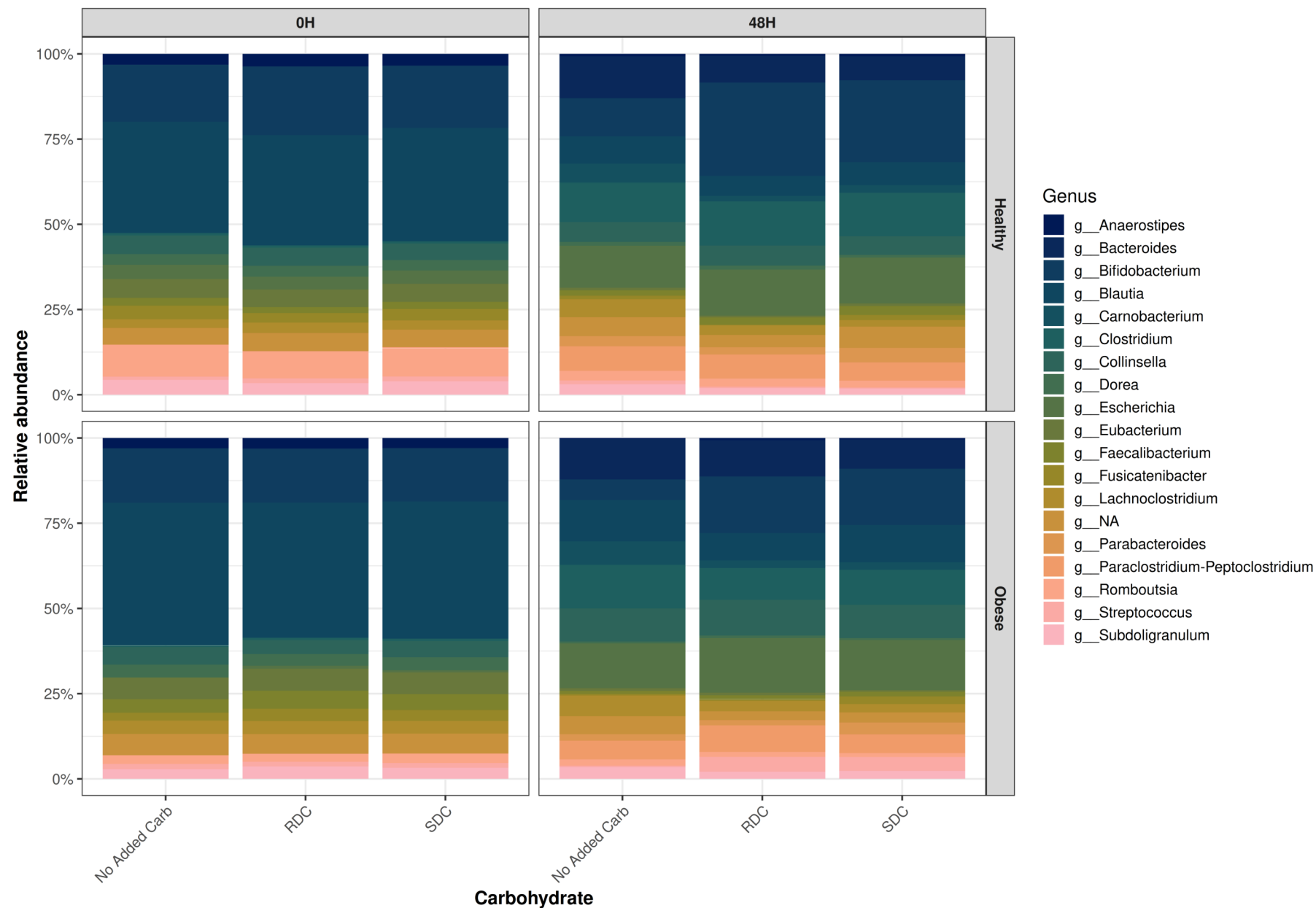
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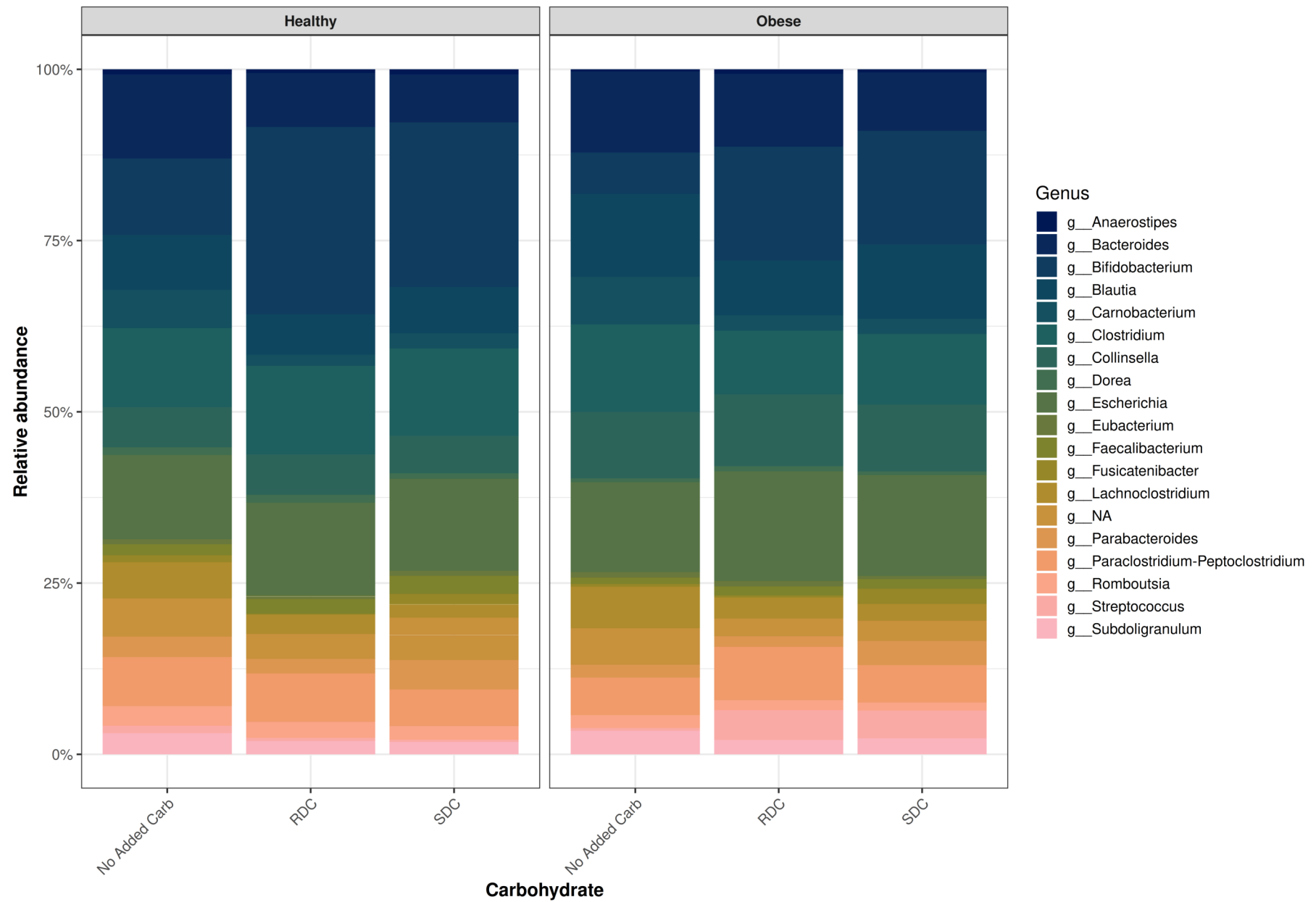
6.4 Community Composition

6.4.1 Genus-Level Aggregation and Overview

Genus-level composition by group, carbohydrate, and timepoint



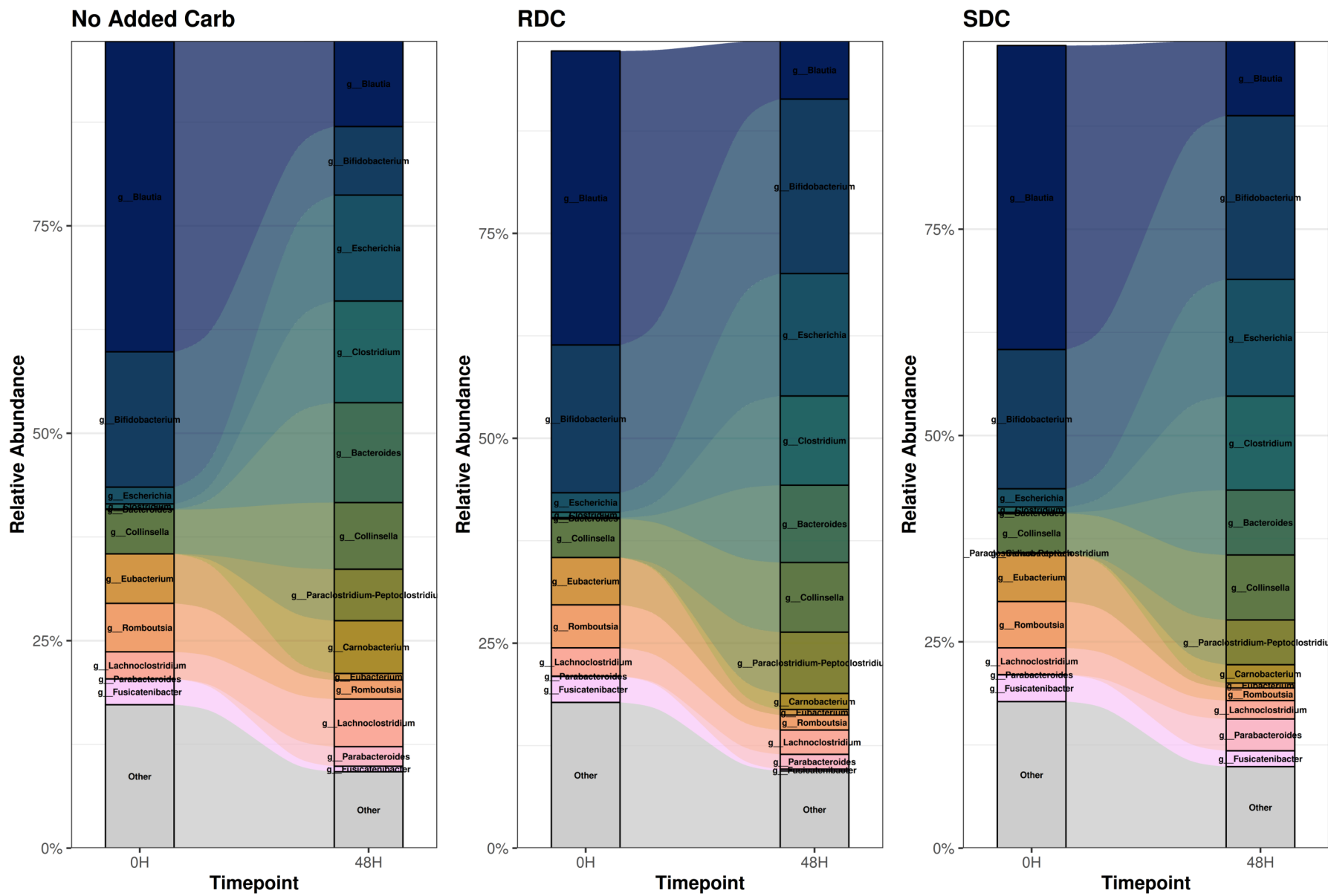
Genus-level composition at 48H



6.4.2 Genus Composition Changes from 0H to 48H by Carbohydrate Type

Genus Composition Changes from 0H to 48H by Carbohydrate Type

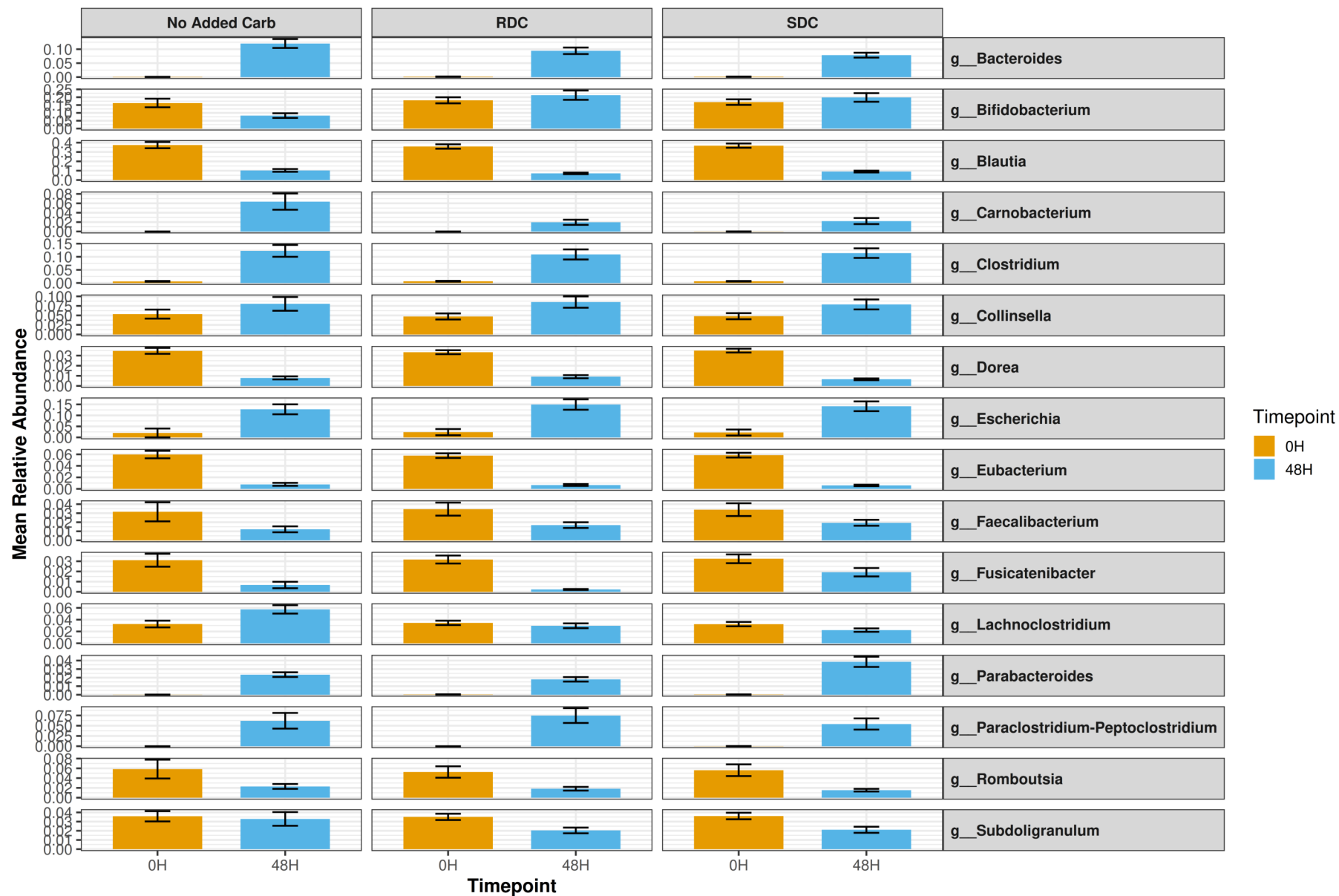
Flow diagram showing relative abundance changes (top 13 genera including DA-important taxa)



6.4.3 Validation: Bar Plots of Genus Abundances by Timepoint and Carbohydrate

Validation: Genus Abundance by Timepoint and Carbohydrate Type

Mean relative abundance $\pm$ SE for top 16 genera (including DA-important taxa)



6.5 Differential Abundance and Predictive Modeling

Differential abundance analysis identifies microbial taxa that differ significantly between experimental conditions. We applied multiple complementary methods to ensure robust identification of differentially abundant taxa: ANCOM-BC2 (Analysis of Compositions of Microbiomes with Bias Correction), which accounts for compositional bias and uses mixed-effects models; MaAsLin2 (Microbiome Multivariable Associations with Linear Models), which handles multivariate associations with normalization and transformations; and LEfSe (Linear Discriminant Analysis Effect Size), which identifies biomarker taxa between groups.

6.5.1 ANCOM-BC2 Analysis

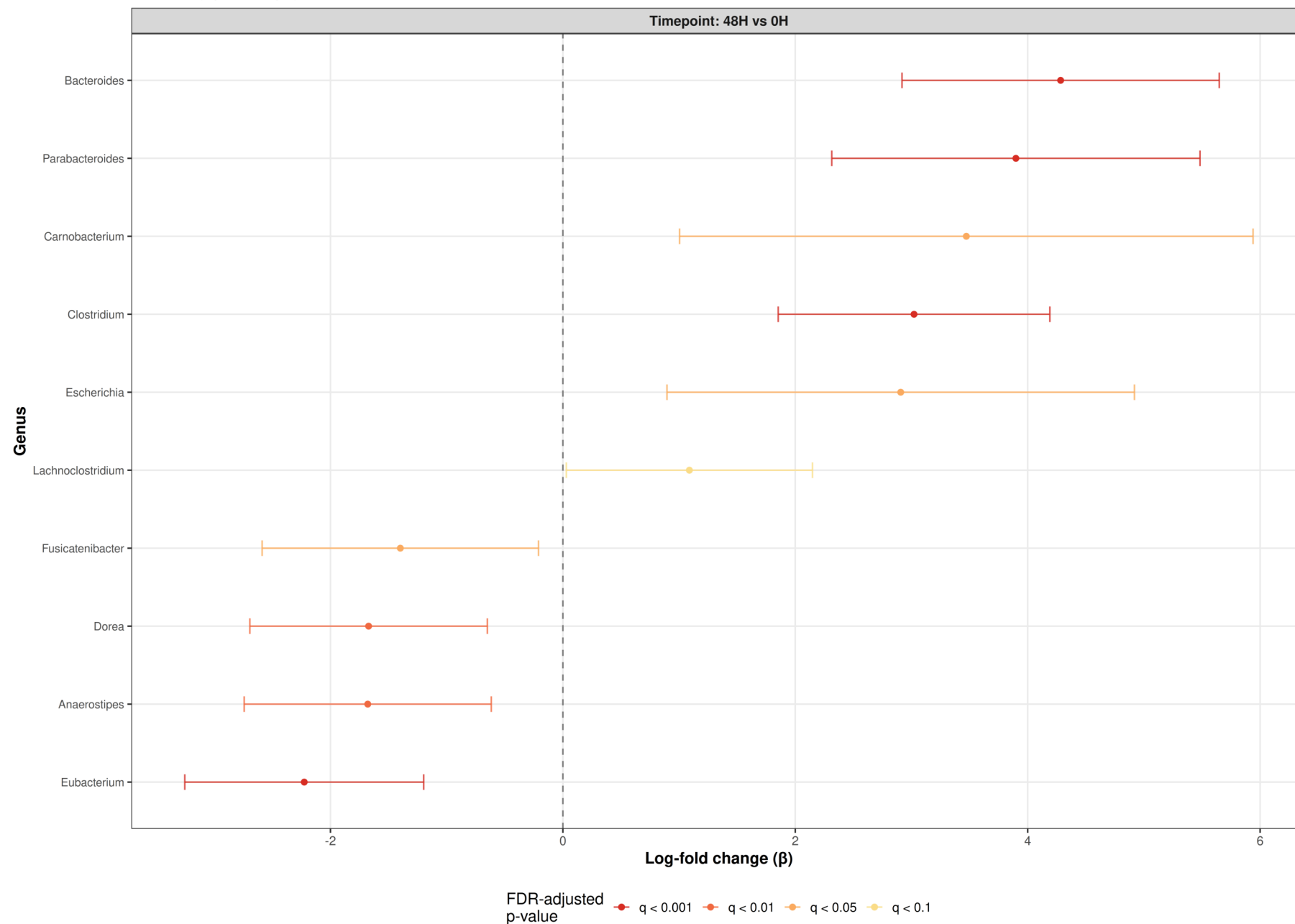
ANCOM-BC2 models microbial abundance using a generalized linear mixed-effects model with bias correction for compositional data. We tested the main effects of group (Healthy vs Obese), carbohydrate type (RDC vs No Added Carb, SDC vs No Added Carb), timepoint (48H vs 0H), and their interactions.

```
## ANCOM-BC2 significant results (q < 0.1): 10 taxon-effect combinations
```

6.5.2 ANCOM-BC2 Forest Plots

ANCOM-BC2 Results: Forest Plot of Significant Taxa

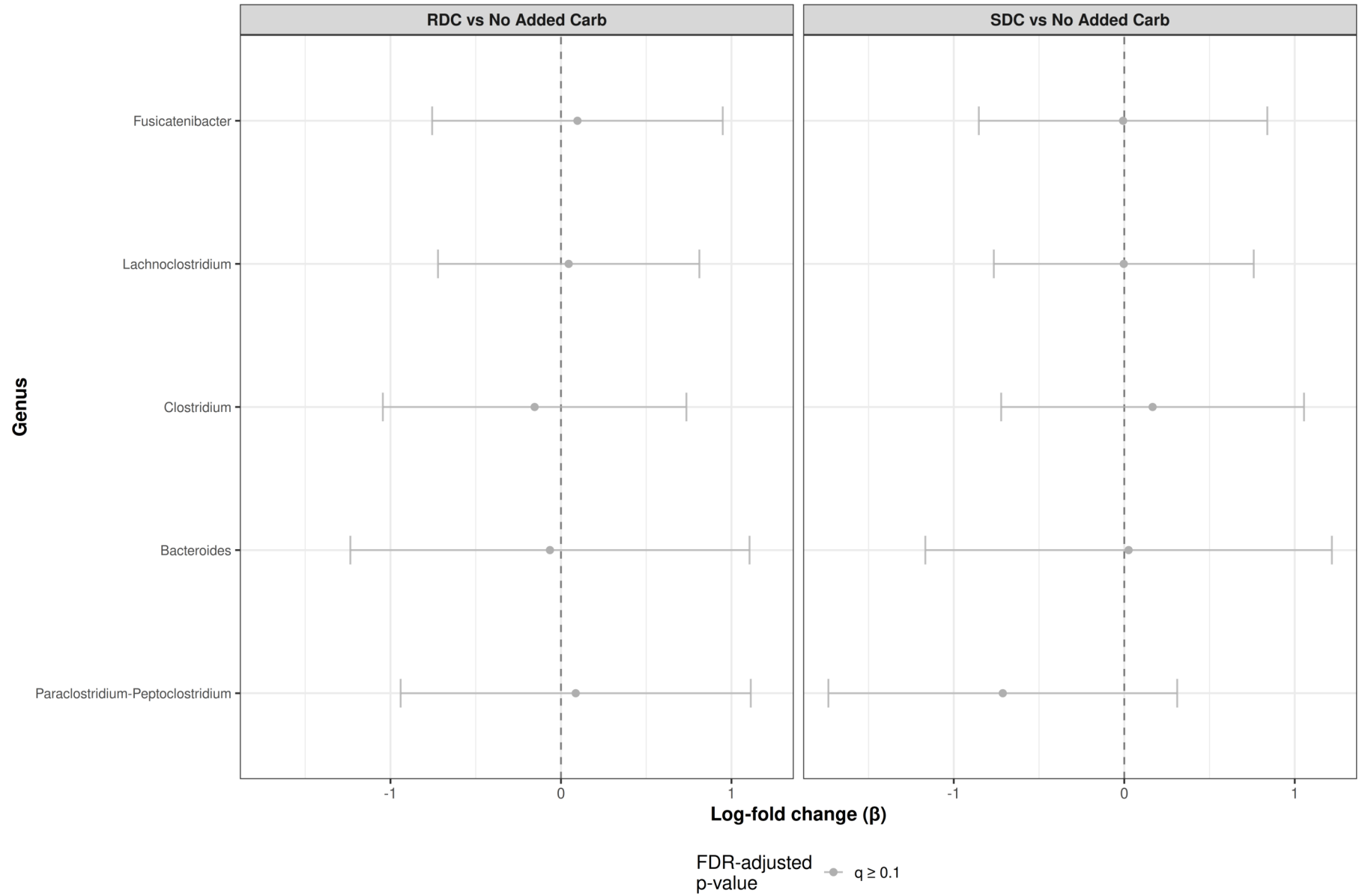
Effect sizes (log-fold changes) with 95% confidence intervals



6.5.3 ANCOM-BC2 Side-by-Side Comparison: Carbohydrate Effects

ANCOM-BC2: Side-by-Side Comparison of Carbohydrate Effects

RDC vs No Added Carb and SDC vs No Added Carb



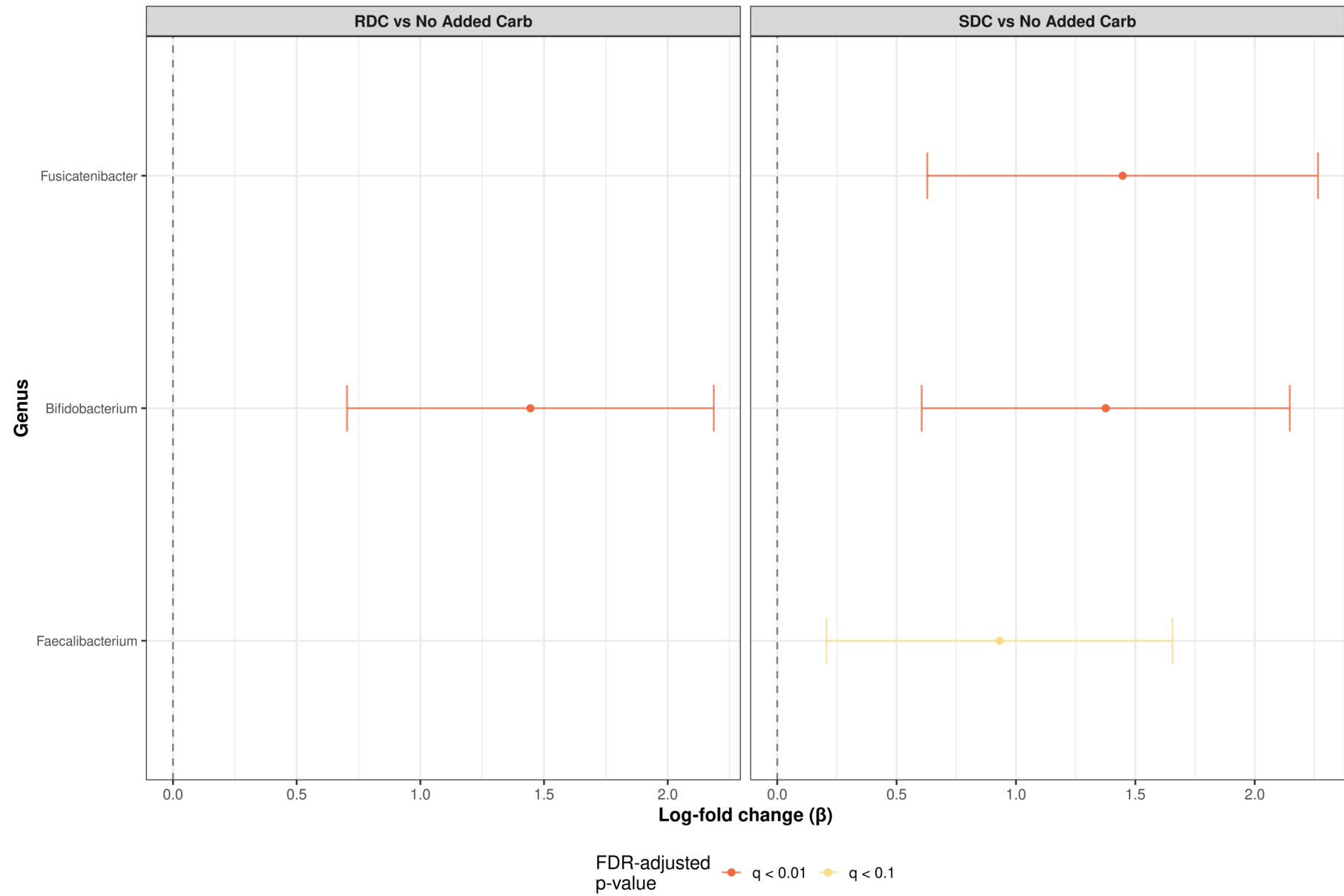
6.5.4 ANCOM-BC2 48H Timepoint Analysis

```
## ANCOM-BC2 48H significant results (q < 0.1): 4 taxon-effect combinations
```

6.5.5 ANCOM-BC2 48H Forest Plots

ANCOM-BC2 Results at 48H: Forest Plot of Significant Taxa

Effect sizes (log-fold changes) with 95% confidence intervals



6.5.6 ANCOM-BC2 RDC vs SDC at 48H

To directly compare the effects of RDC versus SDC at the 48H timepoint, we performed a focused ANCOM-BC2 analysis including only RDC and SDC samples at 48H, excluding the No Added Carb condition.

```
## 48H RDC vs SDC samples for ANCOM-BC2: 126
```

```
## Sample distribution by carbohydrate:
```

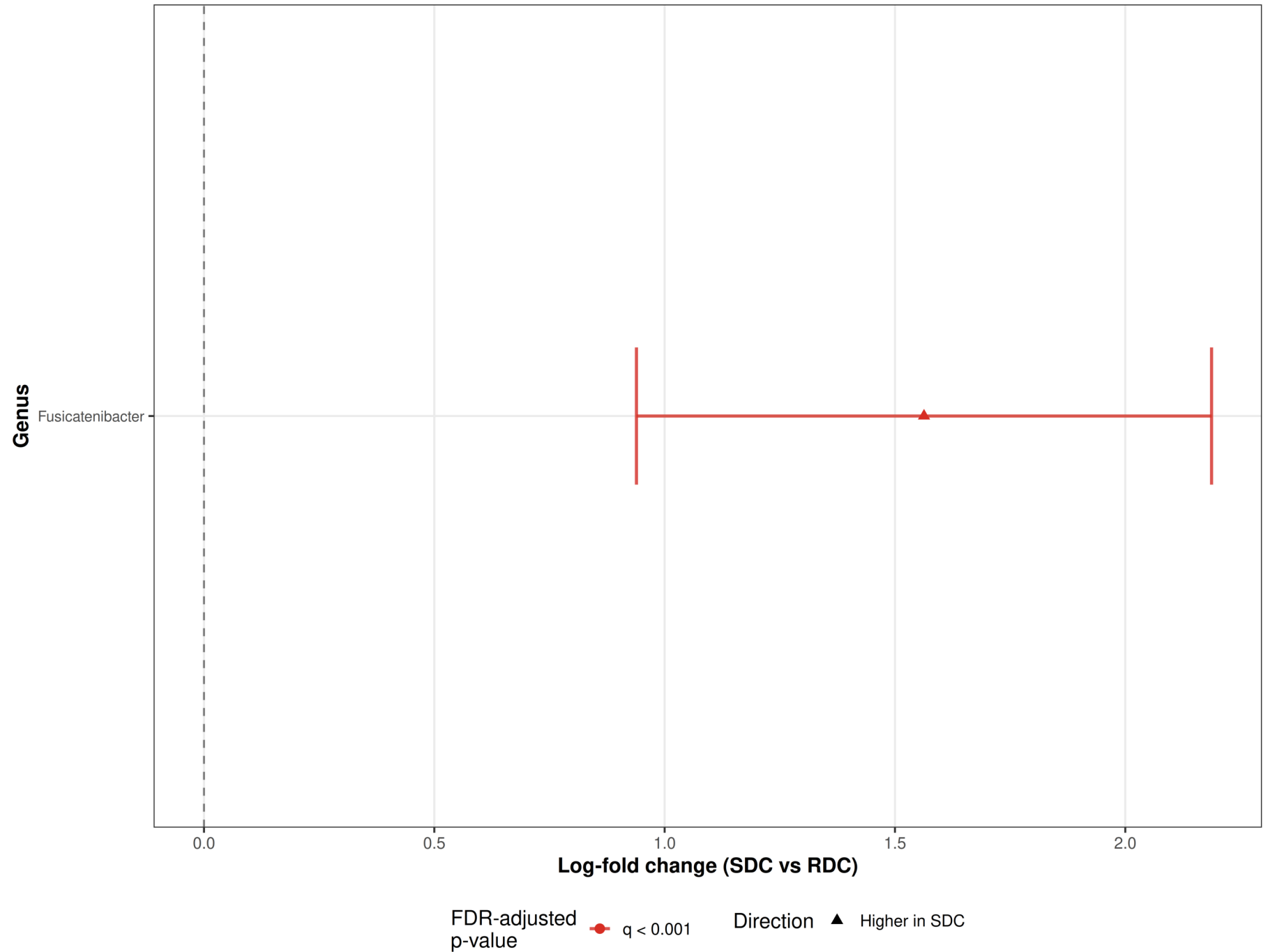
```
##  
## RDC SDC  
## 62 64
```

```
## ANCOM-BC2 48H RDC vs SDC significant results (q < 0.1): 1 taxa
```

6.5.7 Forest Plot: RDC vs SDC at 48H

ANCOM-BC2: RDC vs SDC at 48H

Effect sizes (log-fold change: SDC vs RDC) with 95% confidence intervals



```
##
## Summary of ANCOM-BC2 RDC vs SDC at 48H:
## Total taxa tested: 20
## Significant taxa (q < 0.1): 1
## Higher in SDC (q < 0.1): 1
## Higher in RDC (q < 0.1): 0
```

6.5.8 MaAsLin2 Analysis

MaAsLin2 (Microbiome Multivariable Associations with Linear Models) uses generalized linear models to identify associations between microbial taxa and metadata variables while accounting for confounding factors. We applied MaAsLin2 to identify taxa associated with group, carbohydrate type, and timepoint.

```
## MaAsLin2 significant results (q < 0.1): 32 feature-metadata combinations
```

6.5.9 MaAsLin2 Summary Table

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Search:

Summary of Differential Abundance Results (q < 0.1)

	taxon	Genus Species	DA Method	Variable	Effect Size (coef)	p value	q value
	.	All	f	All	.	All	All
8	seq1	Blautia	MaAsLin2	timepoint=48H	-2.433	8.231999999999999e-29	9.408e-28
9	seq1	Blautia	MaAsLin2	carbohydrate=RDC	-0.7752	0.004065	0.01414
13	seq10	Clostridium	MaAsLin2	timepoint=48H	3.999	1.205e-27	1.205e-26

14	seq10	Clostridium perfringens	ANCOM-BC2	timepoint48H	3.022	8.788e-7	0.000008788
23	seq13	Fusicatenibacter	MaAsLin2	timepoint=48H	-2.833	3.953e-22	3.514e-21
24	seq13	Fusicatenibacter	MaAsLin2	carbohydrate=SDC	0.9733	0.008001	0.02462
25	seq13	Fusicatenibacter saccharivorans	ANCOM-BC2	timepoint48H	-1.399	0.02206	0.04901
26	seq14	Lachnoclostridium	MaAsLin2	carbohydrate=RDC	-1.235	0.0003249	0.001444
27	seq14	Lachnoclostridium	MaAsLin2	timepoint=48H	-0.8555	0.0005913	0.00249
28	seq14	Lachnoclostridium	MaAsLin2	carbohydrate=SDC	-1.16	0.0007063	0.002699
29	seq14	Lachnoclostridium sp32380-sp32437	ANCOM-BC2	timepoint48H	1.089	0.04496	0.08992
10	seq15	Carnobacterium	MaAsLin2	timepoint=48H	1.923	7.530000000000001e-20	6.024e-19
11	seq15	Carnobacterium	MaAsLin2	carbohydrate=RDC	-0.5482	0.0393	0.09824
12	seq15	Carnobacterium mobile	ANCOM-BC2	timepoint48H	3.472	0.008223	0.02056
16	seq17	Dorea	MaAsLin2	timepoint=48H	-3.314	6.163e-33	8.217000000000001e-32
17	seq17	Dorea	MaAsLin2	group=Obese	-0.8518	0.02385	0.06581
18	seq17	Dorea longicatena	ANCOM-BC2	timepoint48H	-1.672	0.001529	0.006117
21	seq18	Eubacterium	MaAsLin2	timepoint=48H	-4.179	6.609e-40	1.057e-38

22	seq18	Eubacterium hallii	ANCOM-BC2	timepoint48H	-2.226	0.00003277	0.0001639
19	seq2	Escherichia	MaAsLin2	timepoint=48H	6.511	1.861e-52	4.962999999999999e-51
20	seq2	Escherichia coli	ANCOM-BC2	timepoint48H	2.907	0.005243	0.01498
41	seq20	Streptococcus	MaAsLin2	timepoint=48H	-2.048	8.996e-17	5.998e-16
42	seq21	Subdoligranulum	MaAsLin2	timepoint=48H	-0.7719	0.0007085	0.002699
30	seq29	Lachnospiraceae	MaAsLin2	timepoint=48H	-1.65	1.059e-10	6.052e-10
31	seq29	Lachnospiraceae	MaAsLin2	carbohydrate=RDC	-0.9757	0.004255	0.01418

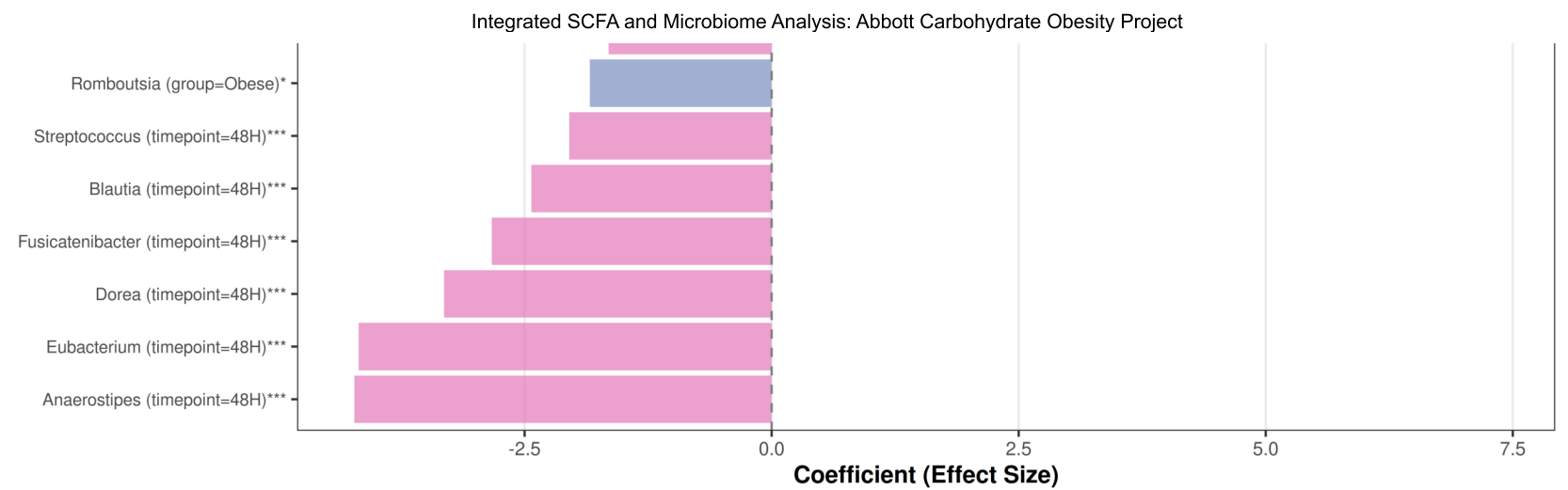
Showing 1 to 25 of 42 entries

[Previous](#)[1](#)[2](#)[Next](#)

6.5.10 MaAsLin2 Visualizations

MaAsLin2 Results: Significant Associations
Coefficient Effect Sizes by Variable



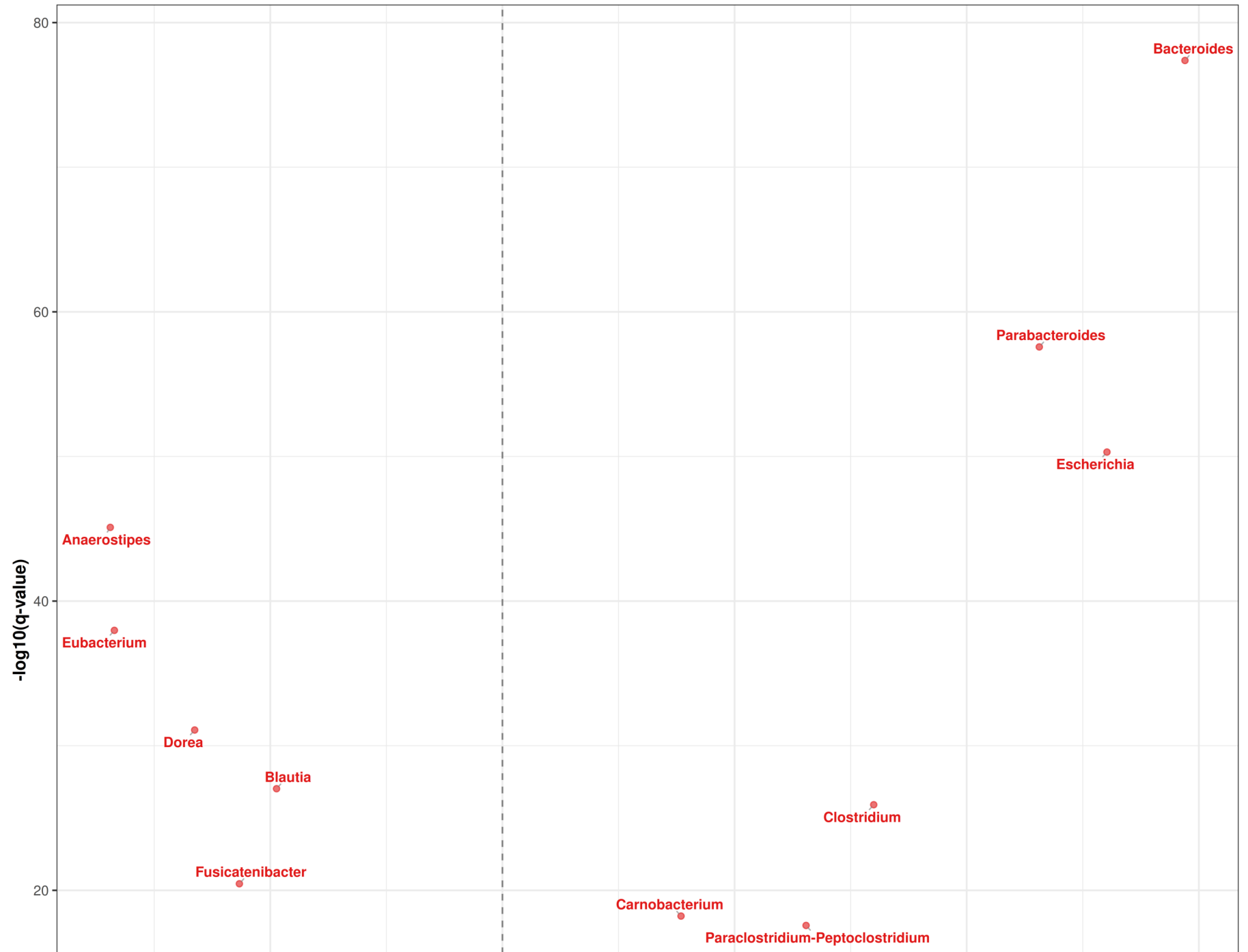


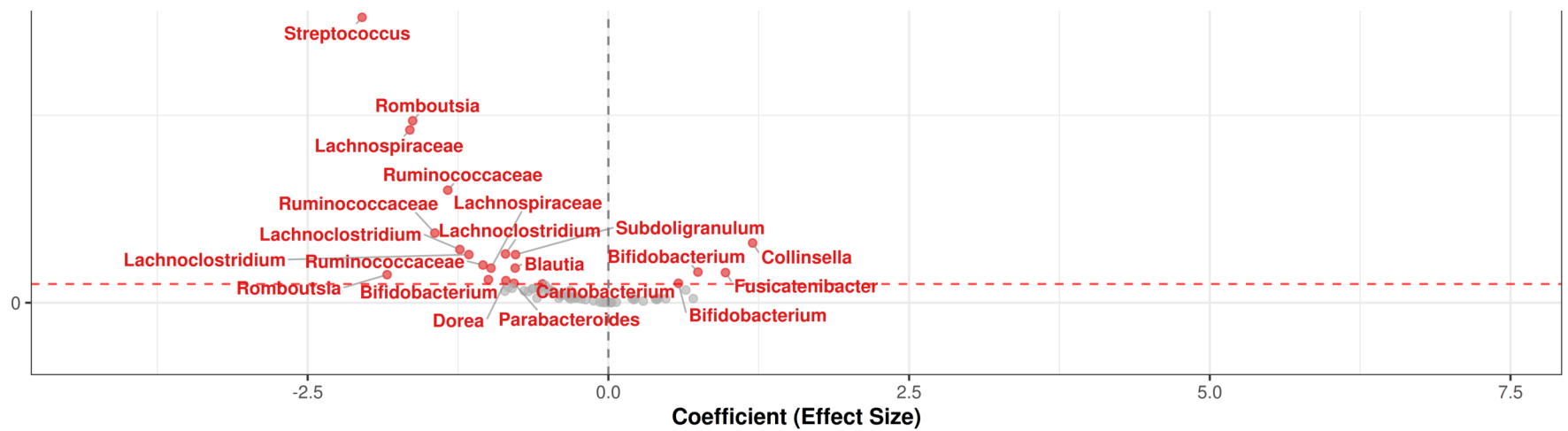
metadata_value carbohydrate=RDC carbohydrate=SDC group=Obese timepoint=48H

*** $q < 0.001$, ** $q < 0.01$, * $q < 0.05$

MaAsLin2 Volcano Plot

Effect Size vs Statistical Significance

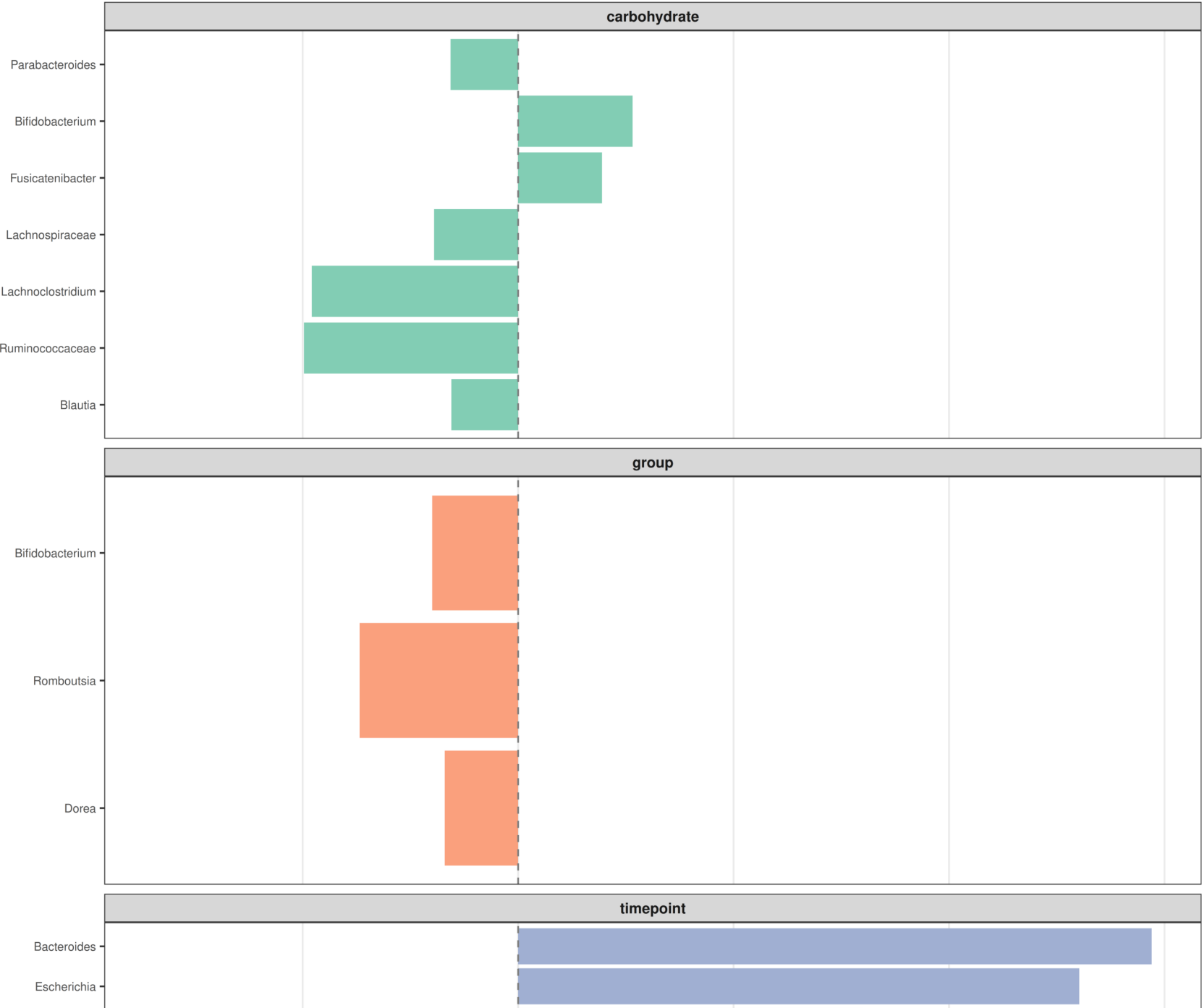


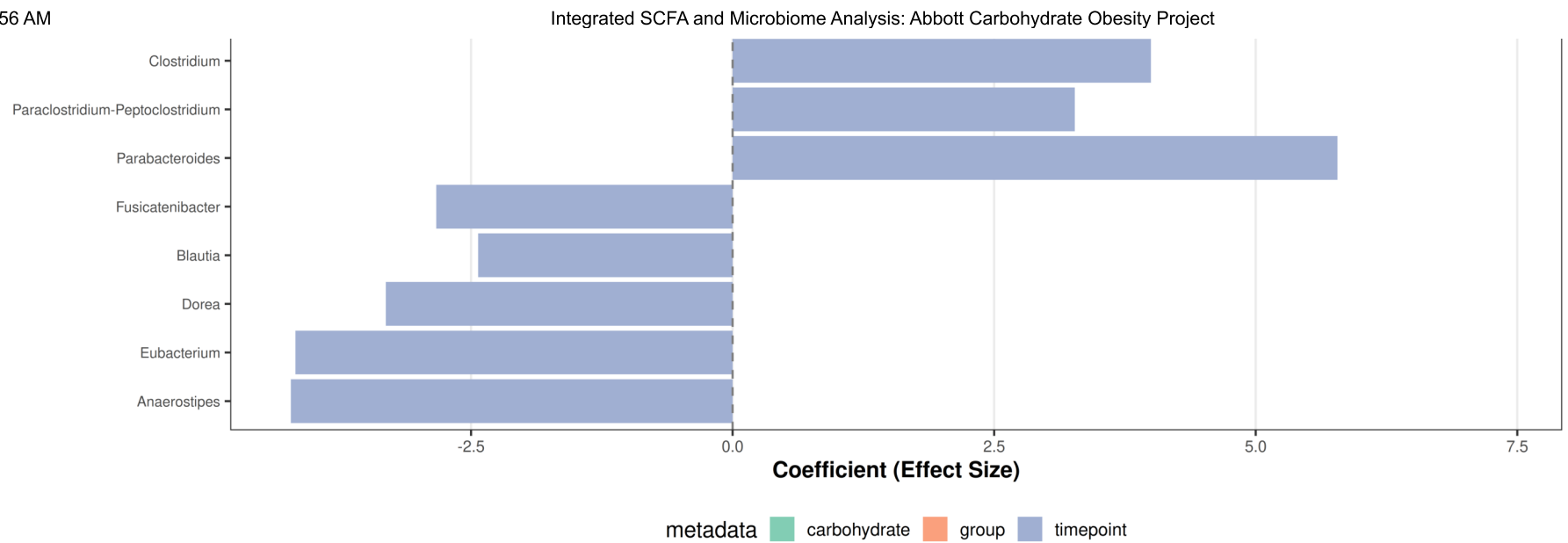


MaAsLin2: Top Effects by Variable

Top 10 largest effects per metadata variable. Comparisons: carbohydrate : RDC vs No Added Carb (positive = more in RDC); SDC vs No

Genus Species





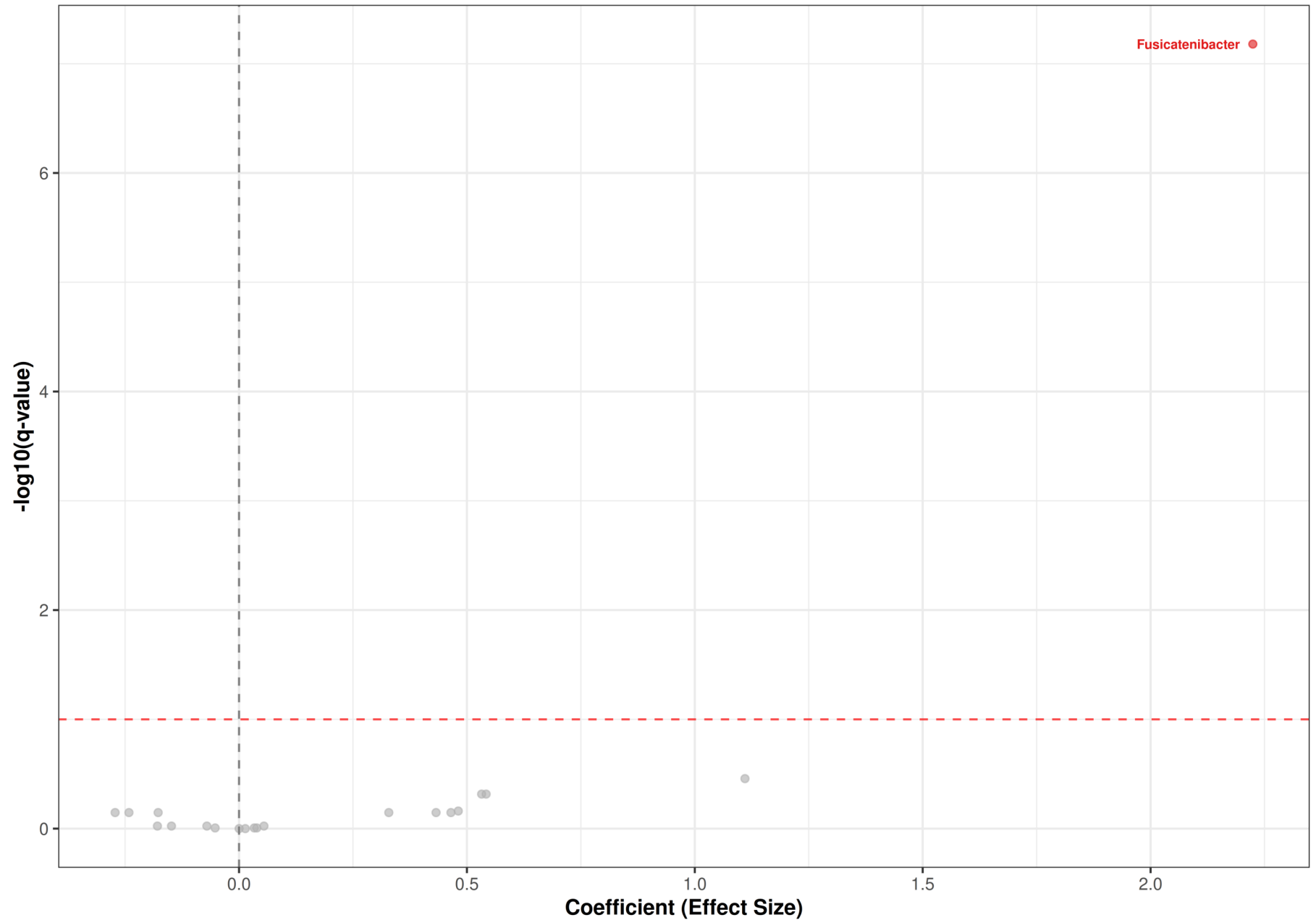
6.5.11 MaAsLin2 48H RDC vs SDC Analysis

```
## MaAsLin2 48H RDC vs SDC significant results (q < 0.1): 1
```

6.5.12 MaAsLin2 48H RDC vs SDC Volcano Plot

MaAsLin2 Volcano Plot: 48H RDC vs SDC

Carbohydrate Effect Size vs Statistical Significance



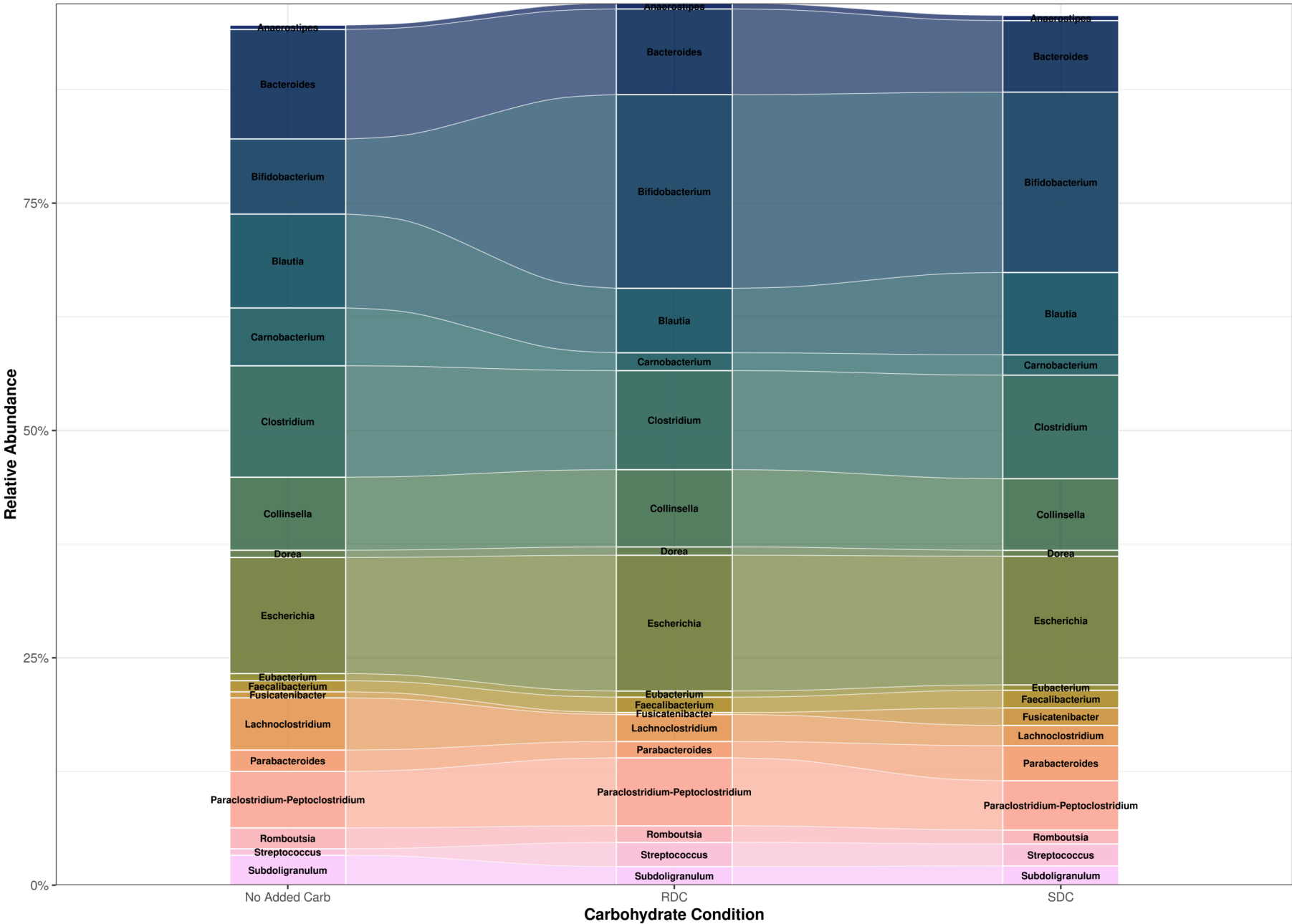
Significant $q \geq 0.1$ $q < 0.1$

6.4.4 Genus Composition at 48H: No Carbohydrate vs RDC vs SDC

This alluvial plot shows the comparison of genus-level composition at 48H across the three carbohydrate conditions (No Added Carb, RDC, and SDC), highlighting the top 25 most abundant genera and including key differentially abundant taxa.

Genus Composition at 48H: Comparison Across Carbohydrate Conditions

Alluvial diagram showing relative abundance for top 19 genera (including DA-important taxa)



```
##  
## 48H Carbohydrate Comparison Summary:
```

```
## - Total genera included: 19
```

```
## - DA-important genera found: 1
```

```
## - DA-important genera: g__Fusicatenibacter
```

6.6 Absolute Abundances

Relative abundance data (compositional) can mask true changes in microbial load. To address this, we integrated qPCR-derived total genome copy estimates per sample to convert relative abundances to absolute abundances (genome copies per sample). This allows us to distinguish between compositional shifts versus true changes in microbial load.

6.6.1 Integration of qPCR Data

```
## qPCR data integration:
```

```
## - Total samples in phyloseq: 292
```

```
## - Samples matched to qPCR data: 292
```

```
## - Match rate: 100 %
```

6.6.2 Conversion to Absolute Abundances

```
## Absolute abundance conversion complete:
```

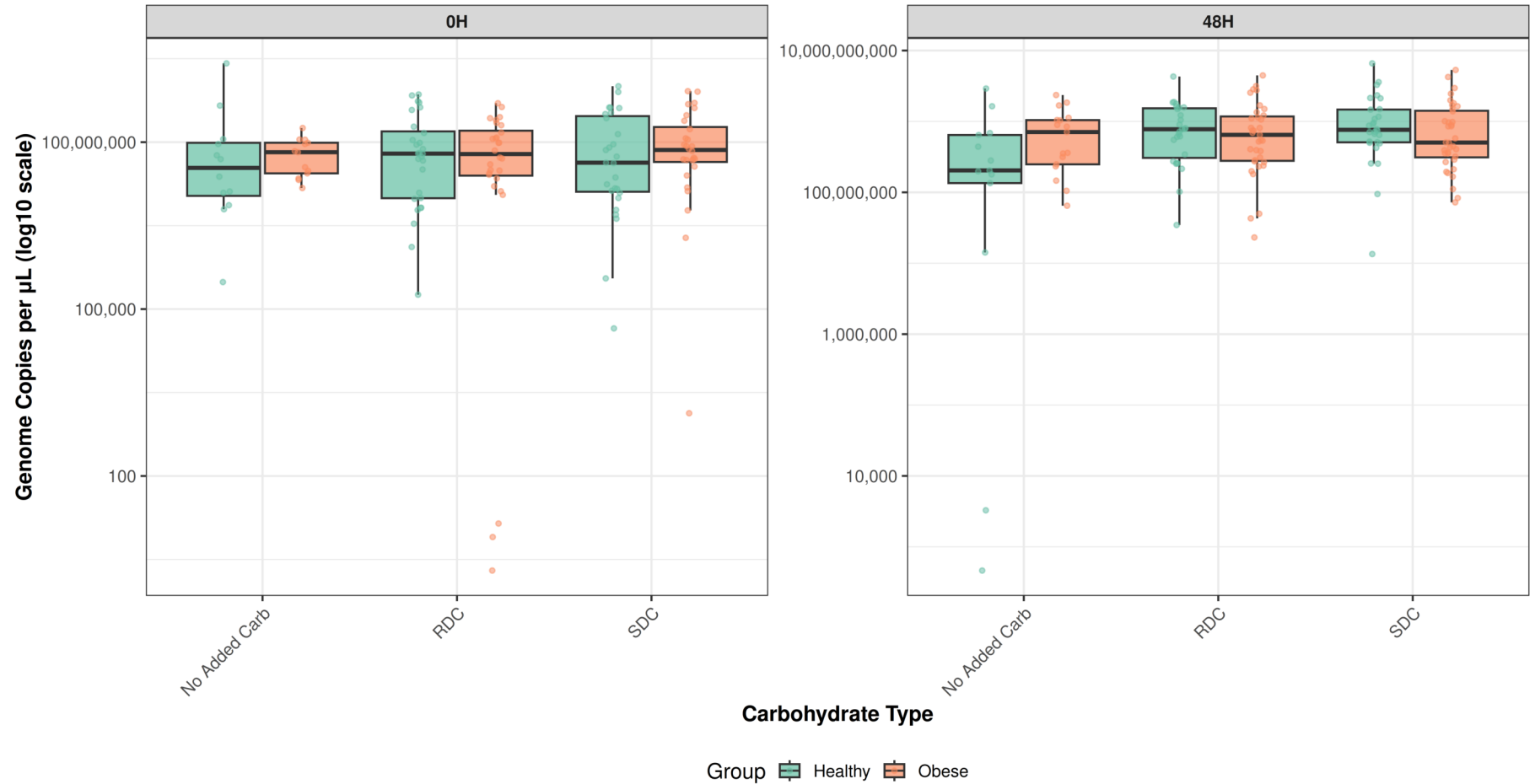
```
## - Samples with qPCR data: 292
```

- Samples without qPCR data: 0

6.6.3 Absolute Abundance Plots by Group, Carbohydrate, and Timepoint

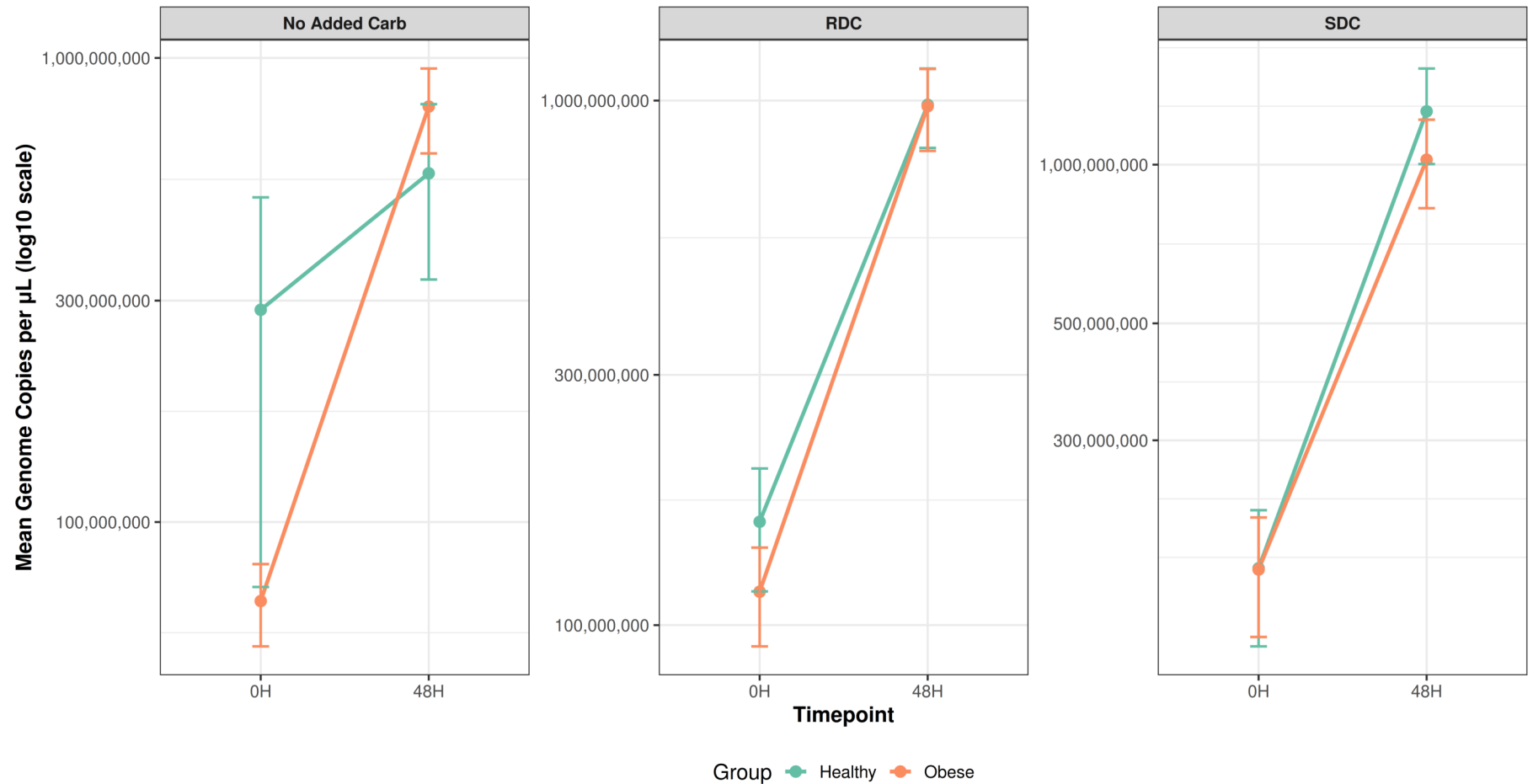
Total Bacterial Genome Copies per μL

qPCR-derived total bacterial load by group, carbohydrate type, and timepoint



Total Bacterial Genome Copies per μL Over Time

Mean $\pm$ SE by carbohydrate type and group



##

qPCR Data Summary:

- Total samples with qPCR data: 292

- Samples by timepoint:

0H 48H
136 156

- Samples by carbohydrate:

##			
##	No Added Carb	RDC	SDC
##	55	118	119

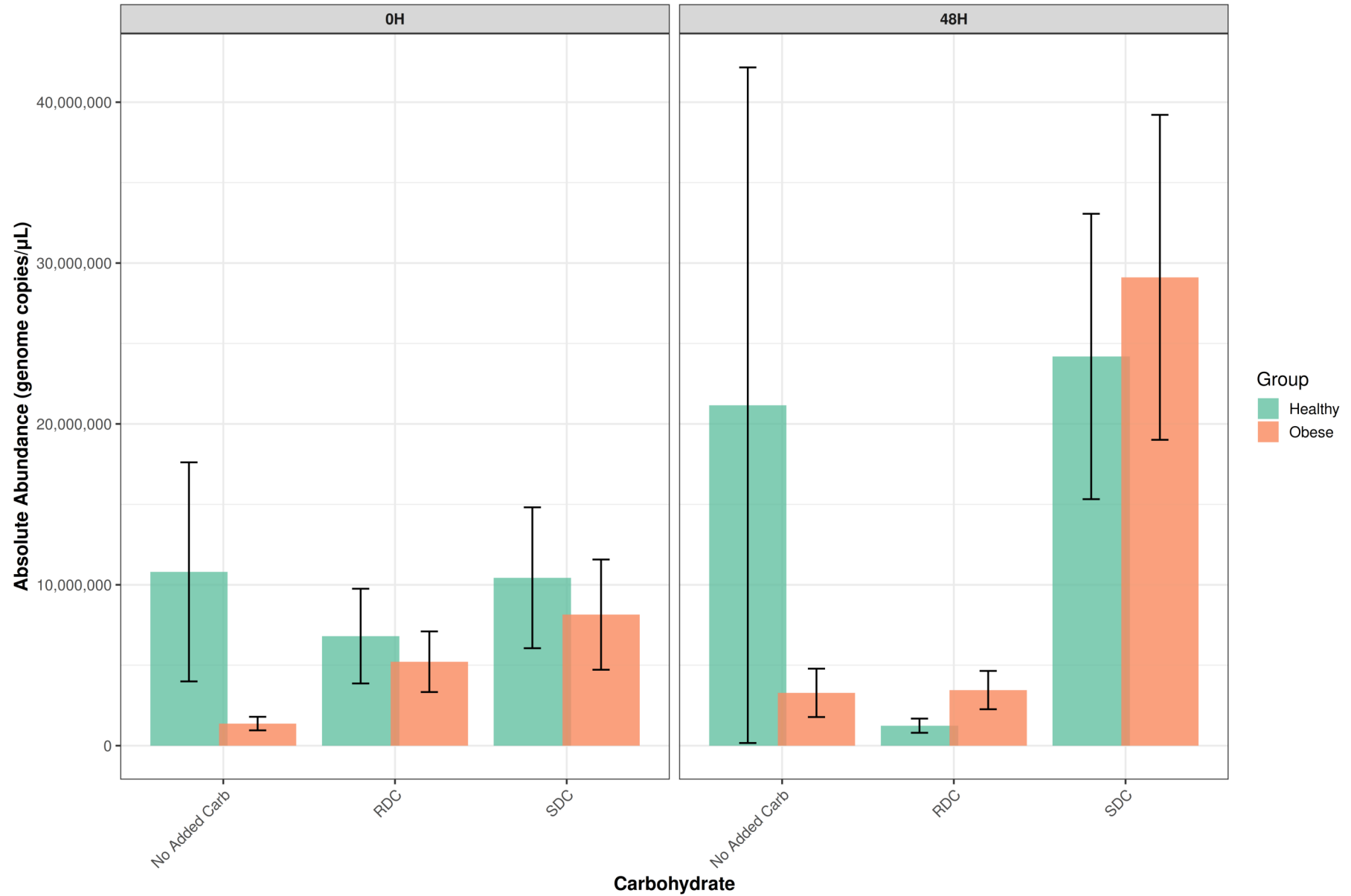
- Samples by group:

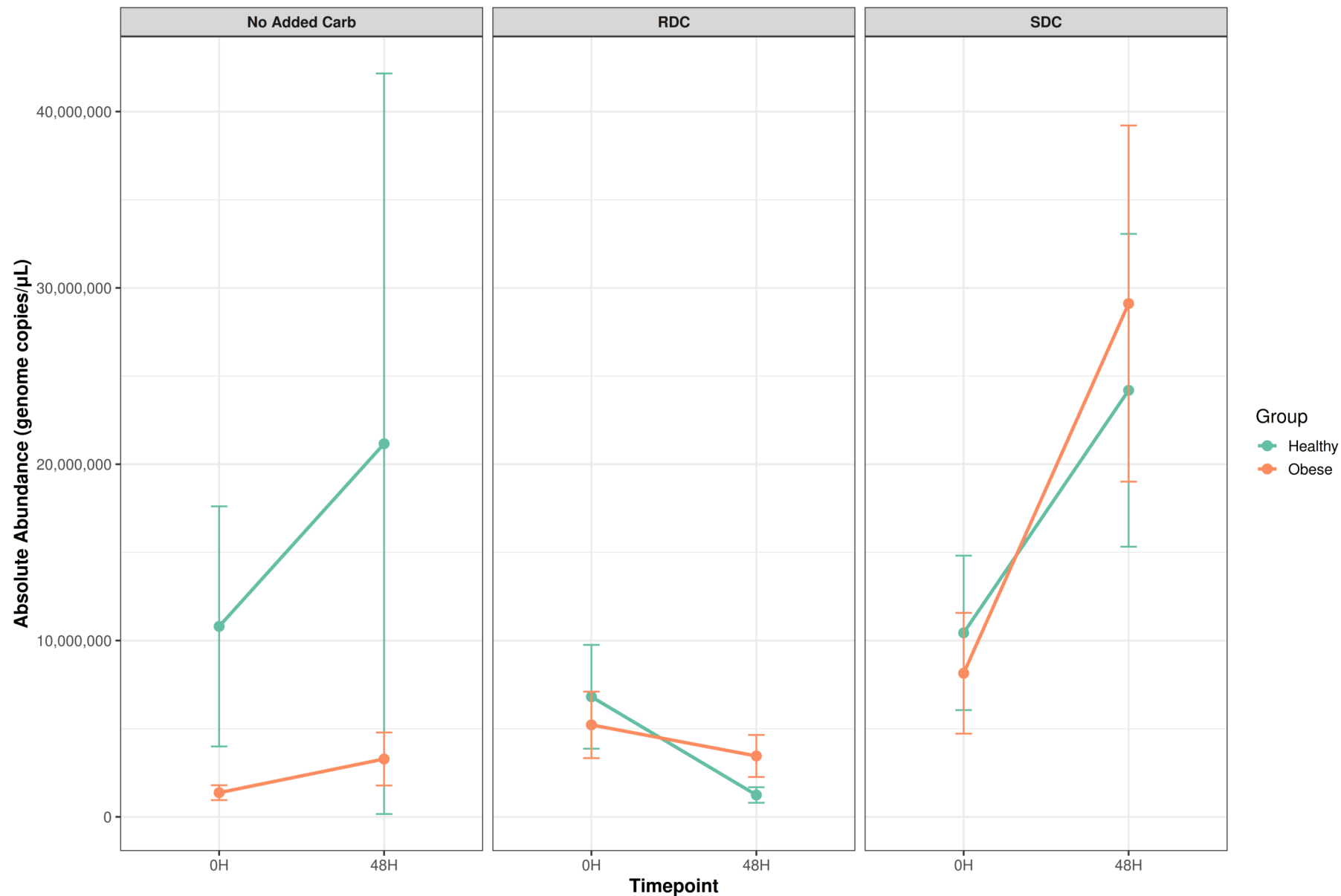
##		
##	Healthy	Obese
##	135	157

6.6.4 Fusicatenibacter Absolute Abundance Plots

g__Fusicatenibacter Absolute Abundance

Mean $\pm$ SE genome copies per μ L by group, carbohydrate, and timepoint



g__Fusicatenibacter Absolute Abundance Over TimeMean $\pm$ SE genome copies per μ L by carbohydrate type and group

7. Integrated Analysis: Taxon–SCFA Relationships

This section integrates SCFA metabolomics with microbiome composition to identify taxa associated with SCFA concentrations. We model associations between microbial taxa and butyrate/propionate concentrations while accounting for carbohydrate type effects, using both relative and absolute abundance data.

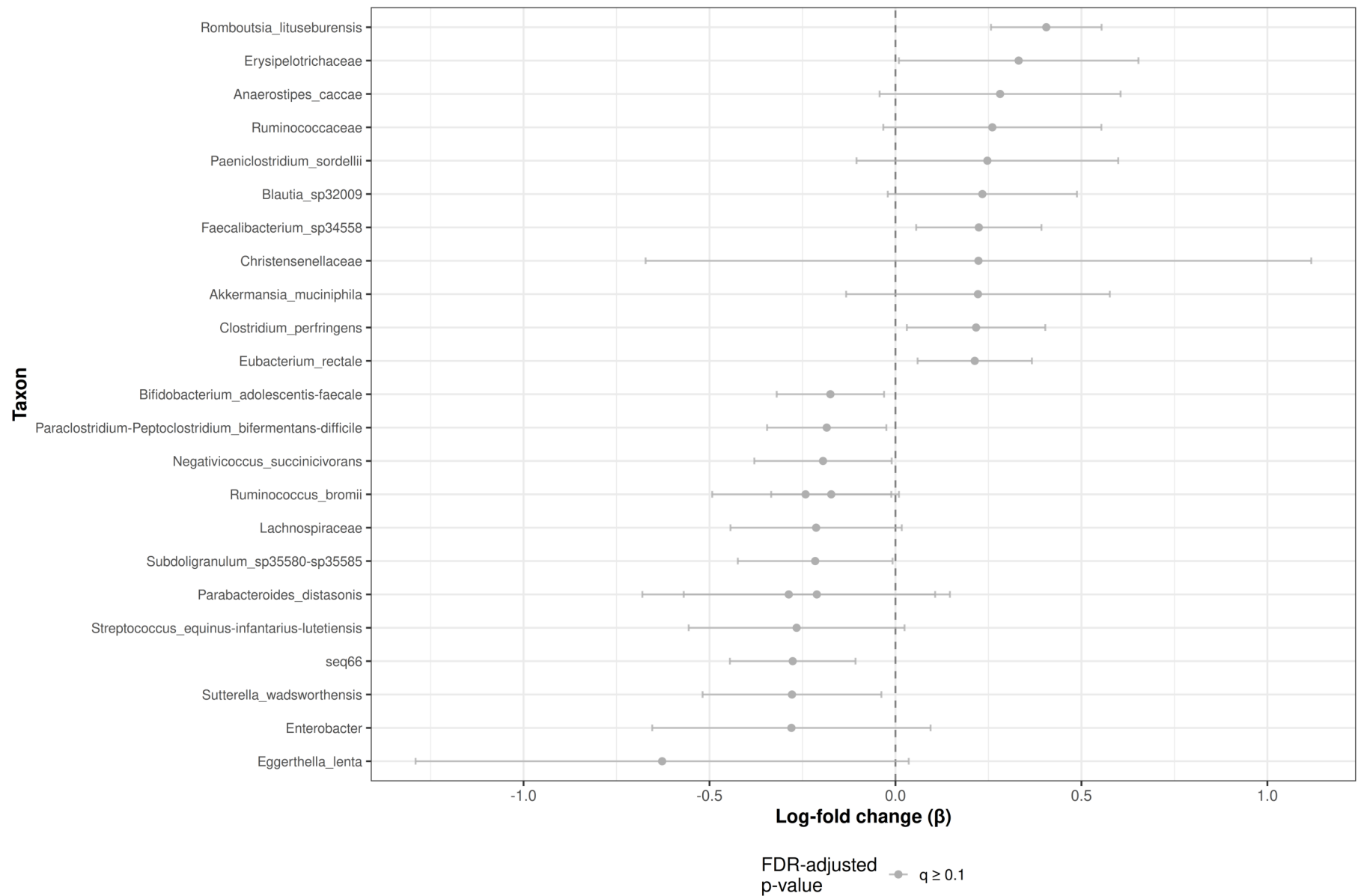
7.1 Integration of SCFA Data with Microbiome Metadata

7.2 Taxon–Butyrate Associations

7.2.1 ANCOM-BC2 and MaAsLin2 Visualizations: Butyrate Associations

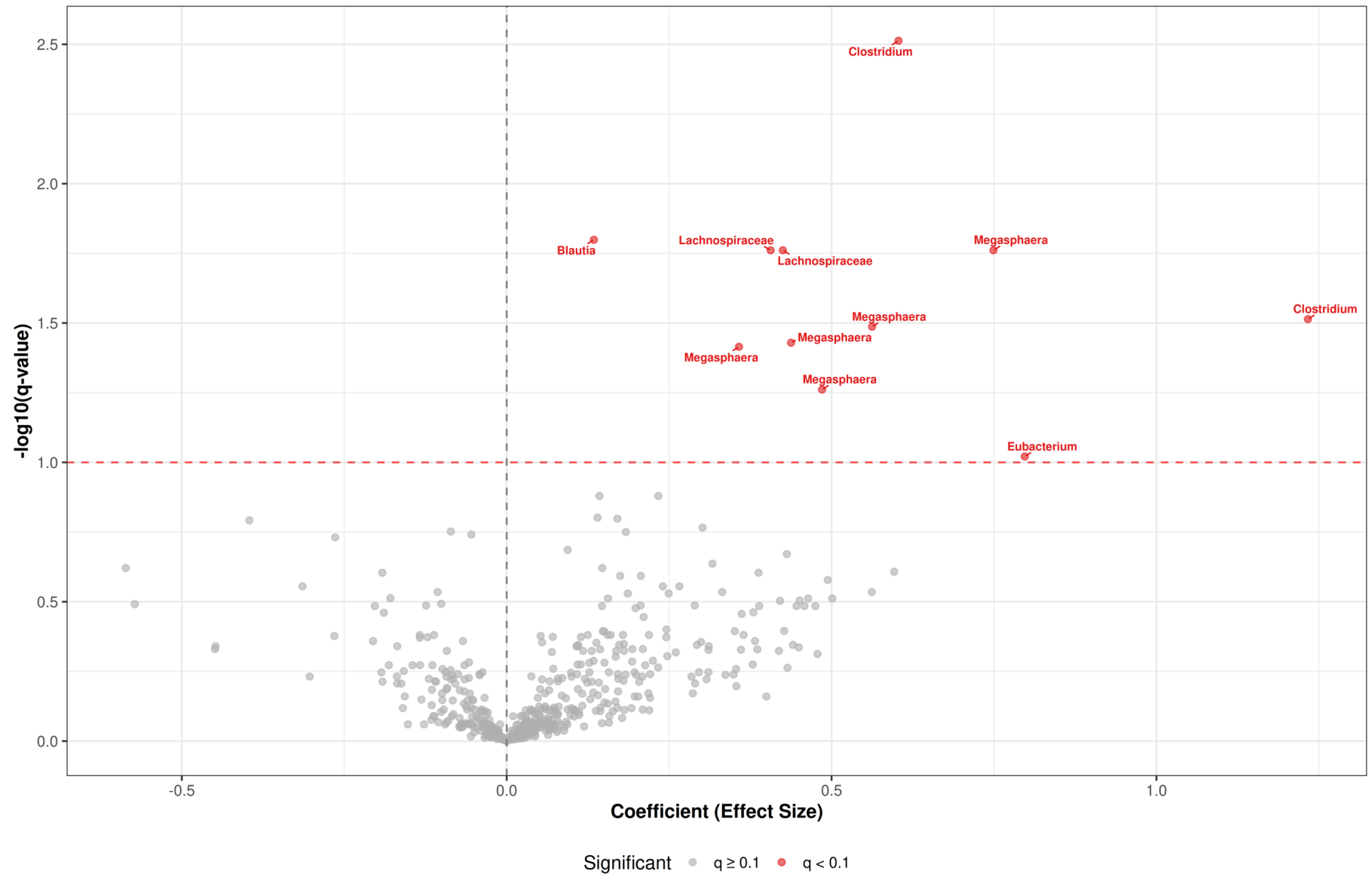
ANCOM-BC2: Taxon Associations with Butyrate

Effect sizes (log-fold change per unit butyrate) with 95% confidence intervals



MaAsLin2: Taxon Associations with Butyrate

Effect Size vs Statistical Significance

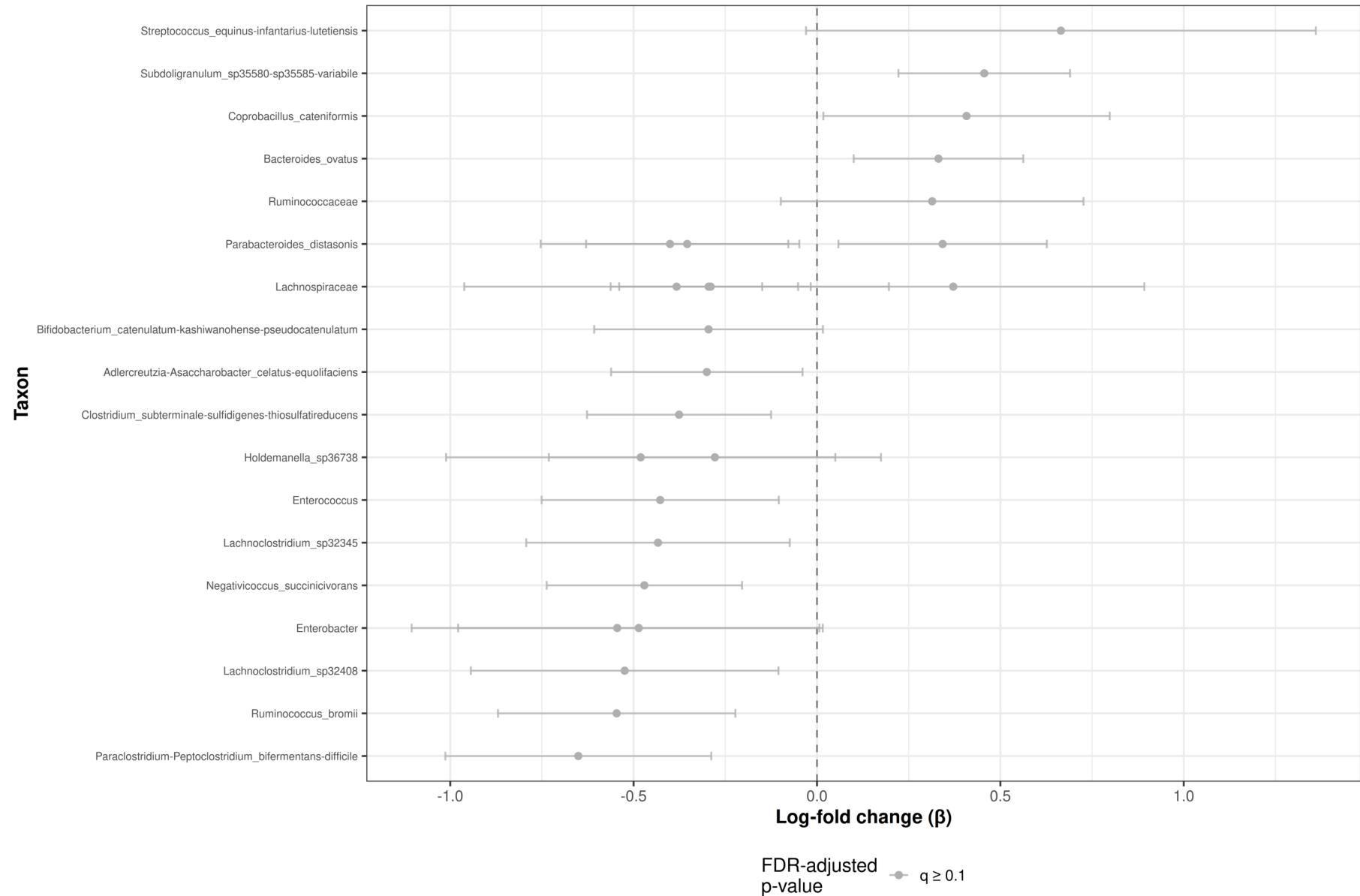
Red dashed line: $q = 0.1$ threshold

7.3 Taxon–Propionate Associations

7.3.1 ANCOM-BC2 and MaAsLin2 Visualizations: Propionate Associations

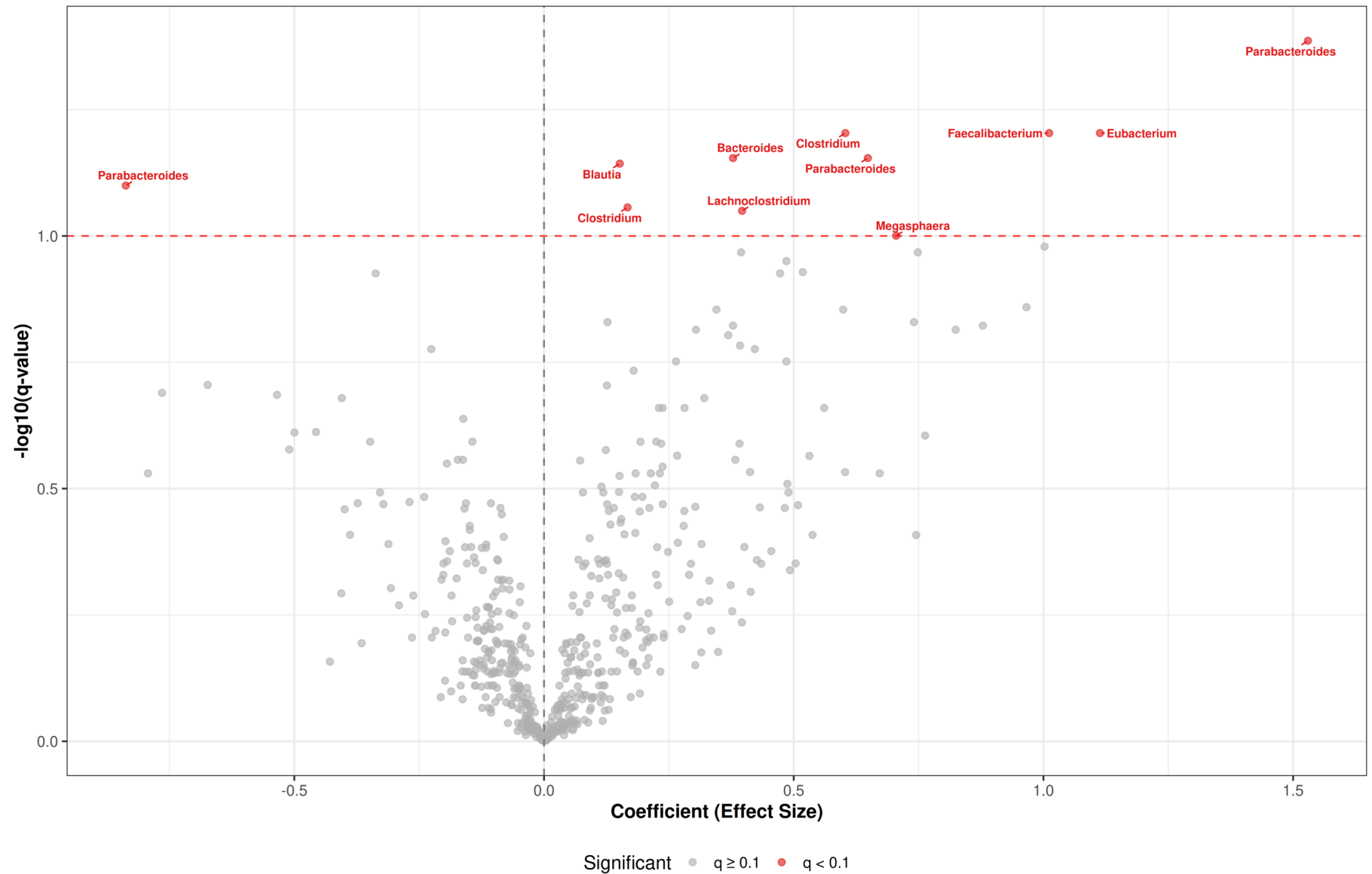
ANCOM-BC2: Taxon Associations with Propionate

Effect sizes (log-fold change per unit propionate) with 95% confidence intervals



MaAsLin2: Taxon Associations with Propionate

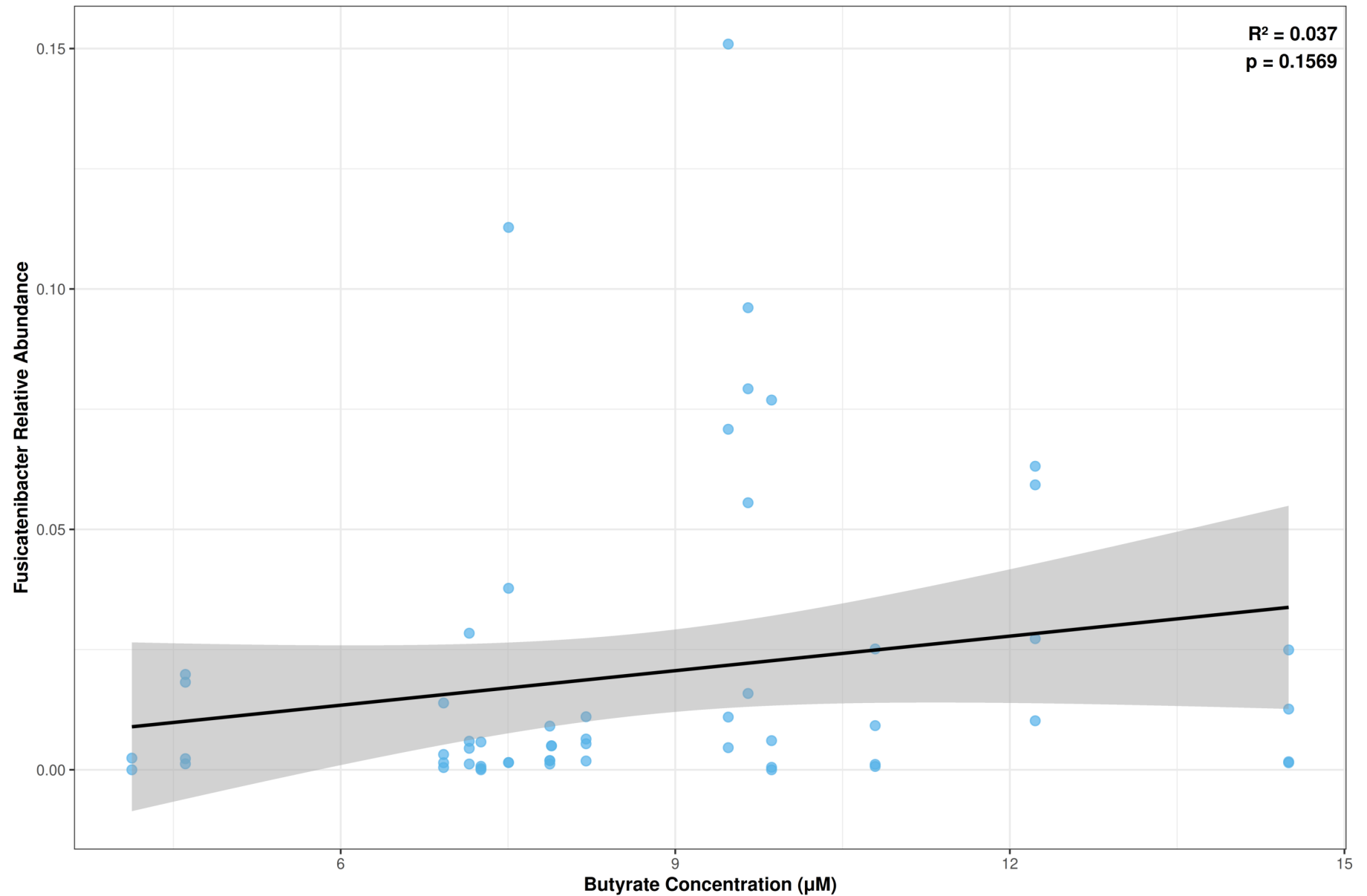
Effect Size vs Statistical Significance

Red dashed line: $q = 0.1$ threshold

7.4 Fuscatenibacter–SCFA Correlations

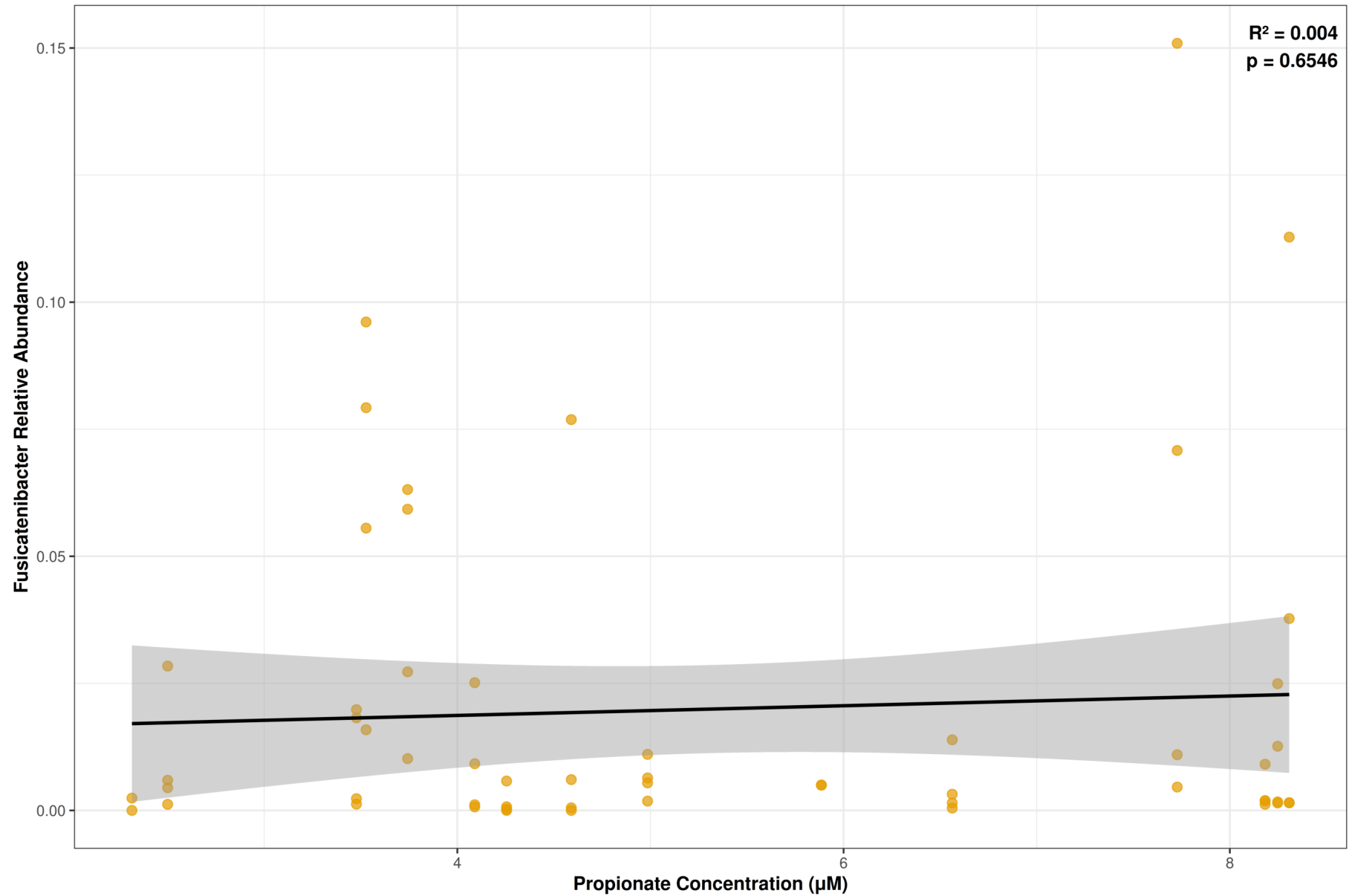
Fuscatenibacter vs Butyrate (Relative Abundance)

48H timepoint, SDC carbohydrate only



Fusicatenibacter vs Propionate (Relative Abundance)

48H timepoint, SDC carbohydrate only



```
##
## Fusicatenibacter-SCFA Correlations (48H, SDC only):
## =====
## Butyrate:
##   R2 = 0.0368, p = 0.1569
##   Spearman rho = 0.287
##   N = 56
##
## Propionate:
##   R2 = 0.0037, p = 0.6546
##   Spearman rho = -0.031
##   N = 56
```

7.5 Responder vs Non-Responder Analysis

```
## Responder Status Overlap:
```

```
## - Total subjects: 48
```

```
##           Propionate
## Butyrate   Non-Responder Responder
## Non-Responder      12      12
## Responder          12      12
## - Both Responder: 12 (25.0%)
## - Both Non-Responder: 12 (25.0%)
```

```
##
## 48H Microbiome Composition Comparison:
```

```
## - Samples with responder status: 136
```

7.6 Microbial Composition Differences: Butyrate Responders vs Non-Responders

To investigate whether individual variation in SCFA response is associated with differences in microbiome composition, we classified subjects as “Responders” or “Non-Responders” based on their change in SCFA concentration from baseline (0H) to 48 hours (48H).

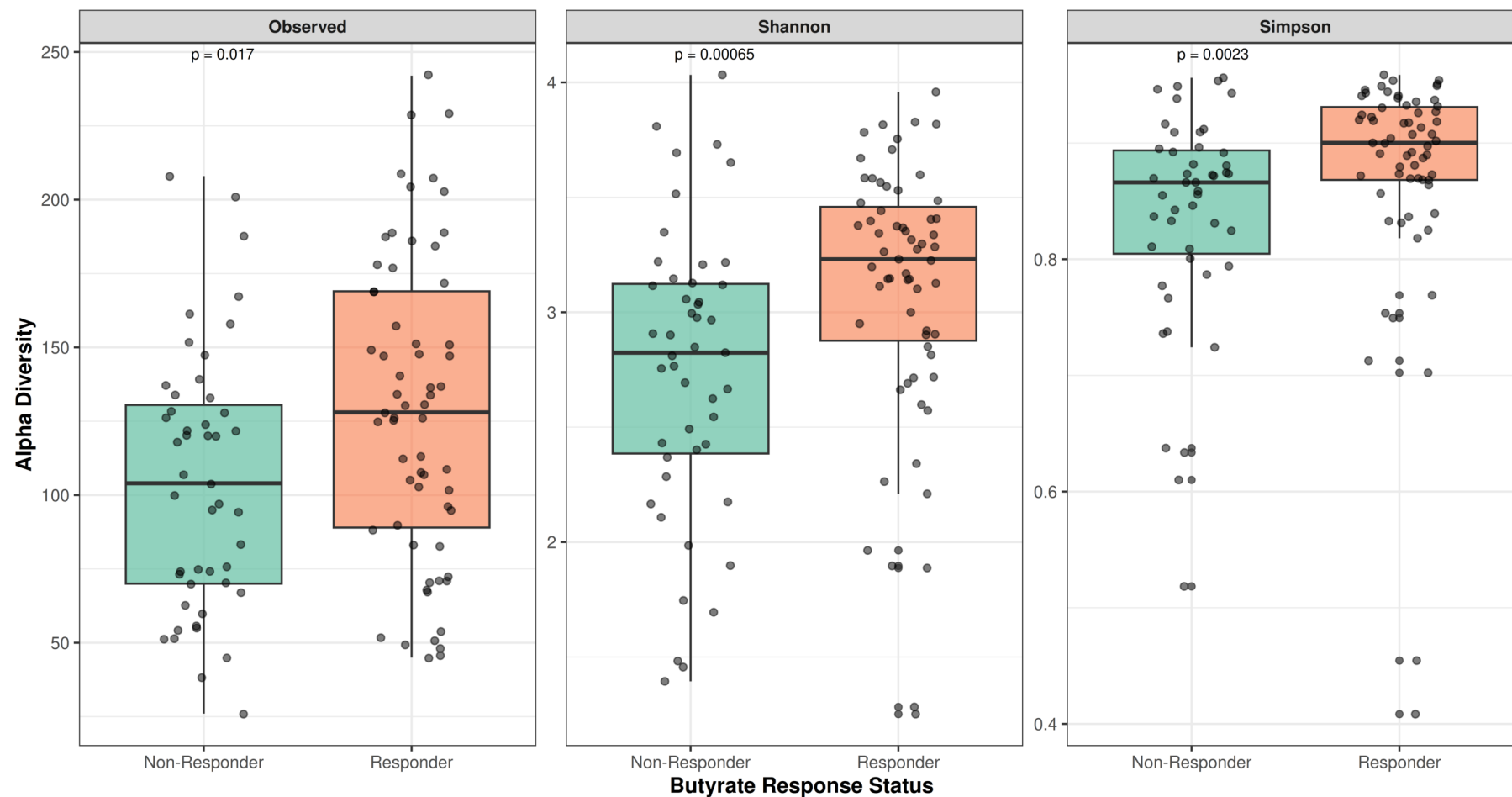
Definition of Responder Status: - For each SCFA (butyrate and propionate) and each carbohydrate condition (RDC and SDC), we calculated the delta change: $\Delta = [\text{SCFA}]_{48\text{h}} - [\text{SCFA}]_{0\text{h}}$ - Subjects were classified separately for each analyte and carbohydrate type using a median split approach - **Responders:** Subjects with $\Delta \geq \text{median } \Delta$ for that specific analyte-carbohydrate combination - **Non-Responders:** Subjects with $\Delta < \text{median } \Delta$ for that specific analyte-carbohydrate combination

This approach allows us to identify whether individuals who show greater increases in SCFA production (responders) have distinct microbial communities compared to those with smaller increases (non-responders), potentially revealing microbiome features associated with enhanced SCFA production capacity.

In this section, we compare the microbial composition (alpha diversity, beta diversity, and community structure) between butyrate responders and non-responders at the 48H timepoint, focusing on samples from RDC and SDC conditions (excluding No Added Carb controls).

Alpha Diversity: Butyrate Responders vs Non-Responders (48H)

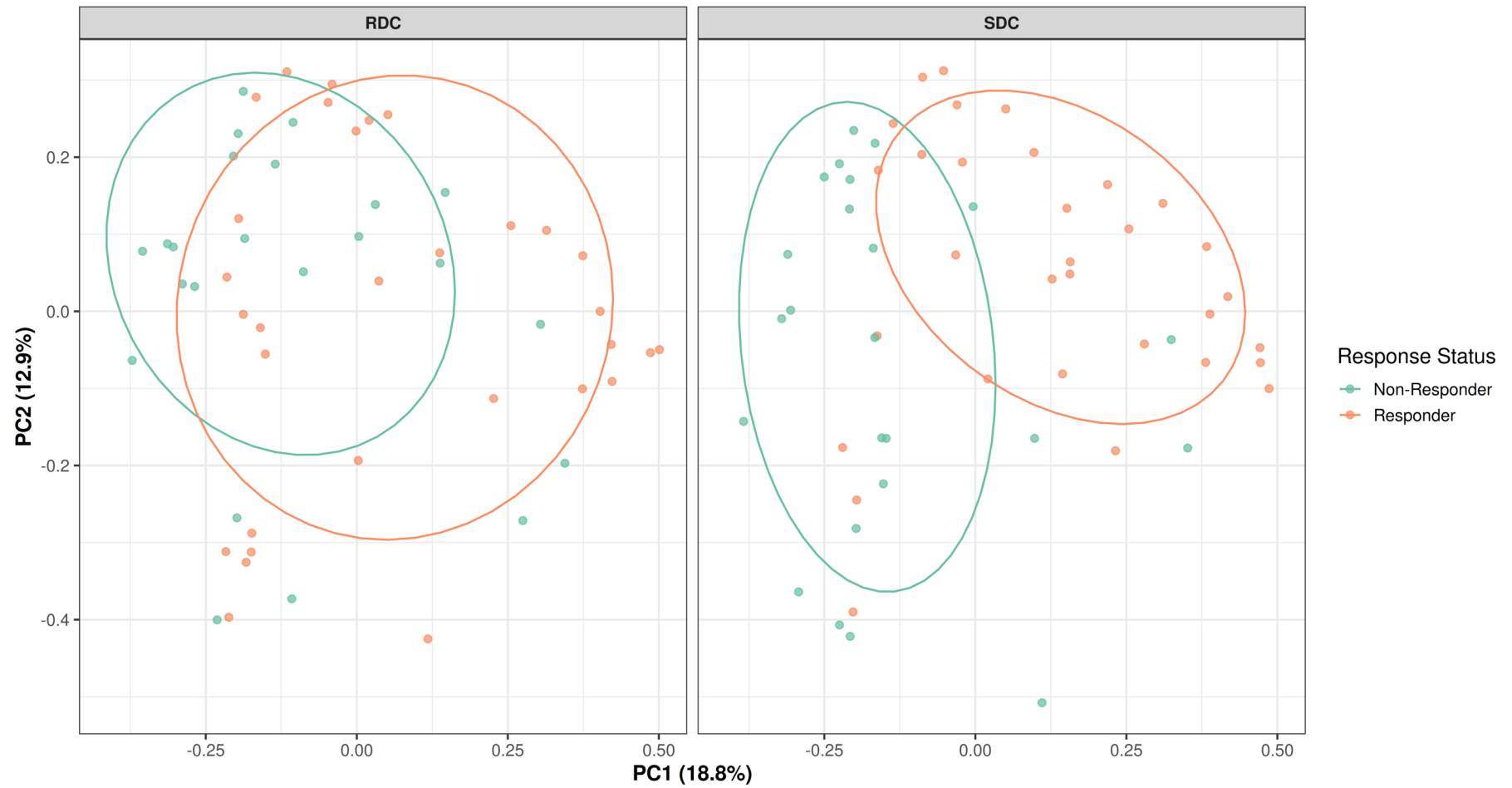
Comparison across RDC and SDC conditions (Wilcoxon test p-values shown)



```
##
## Wilcoxon Test Results: Butyrate Responders vs Non-Responders (Alpha Diversity)
## =====
## # A tibble: 3 × 6
##   Metric   Group1      Group2   `p value` `p adjusted` Significance
##   <chr>    <chr>      <chr>      <dbl>      <dbl> <chr>
## 1 Observed Non-Responder Responder  0.0167      0.0167 *
## 2 Shannon  Non-Responder Responder  0.000655    0.00196 **
## 3 Simpson  Non-Responder Responder  0.00232     0.00348 **
```

Beta Diversity: Butyrate Responders vs Non-Responders (48H)

Bray-Curtis PCoA



```
##
## PERMANOVA: Butyrate Responder Status (48H):
## Permutation test for adonis under reduced model
## Blocks: strata
## Permutation: free
## Number of permutations: 999
##
## vegan::adonis2(formula = dist_bray_resp ~ responder_butyrate * carbohydrate, data = meta_resp, permutations = 999, strata
= meta_resp$subject_id)
##           Df SumOfSqs      R2      F Pr(>F)
## Model      3      2.835 0.08264 3.1832  0.017 *
## Residual 106     31.473 0.91736
## Total    109     34.308 1.00000
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

7.6.1 Differential Abundance Analysis: Butyrate Responders vs Non-Responders

To identify specific taxa that differ between butyrate responders and non-responders, we performed differential abundance analysis using ANCOM-BC2 and MaAsLin2.

```
## ANCOM-BC2 butyrate responder: 0 significant taxa (q < 0.1)
```

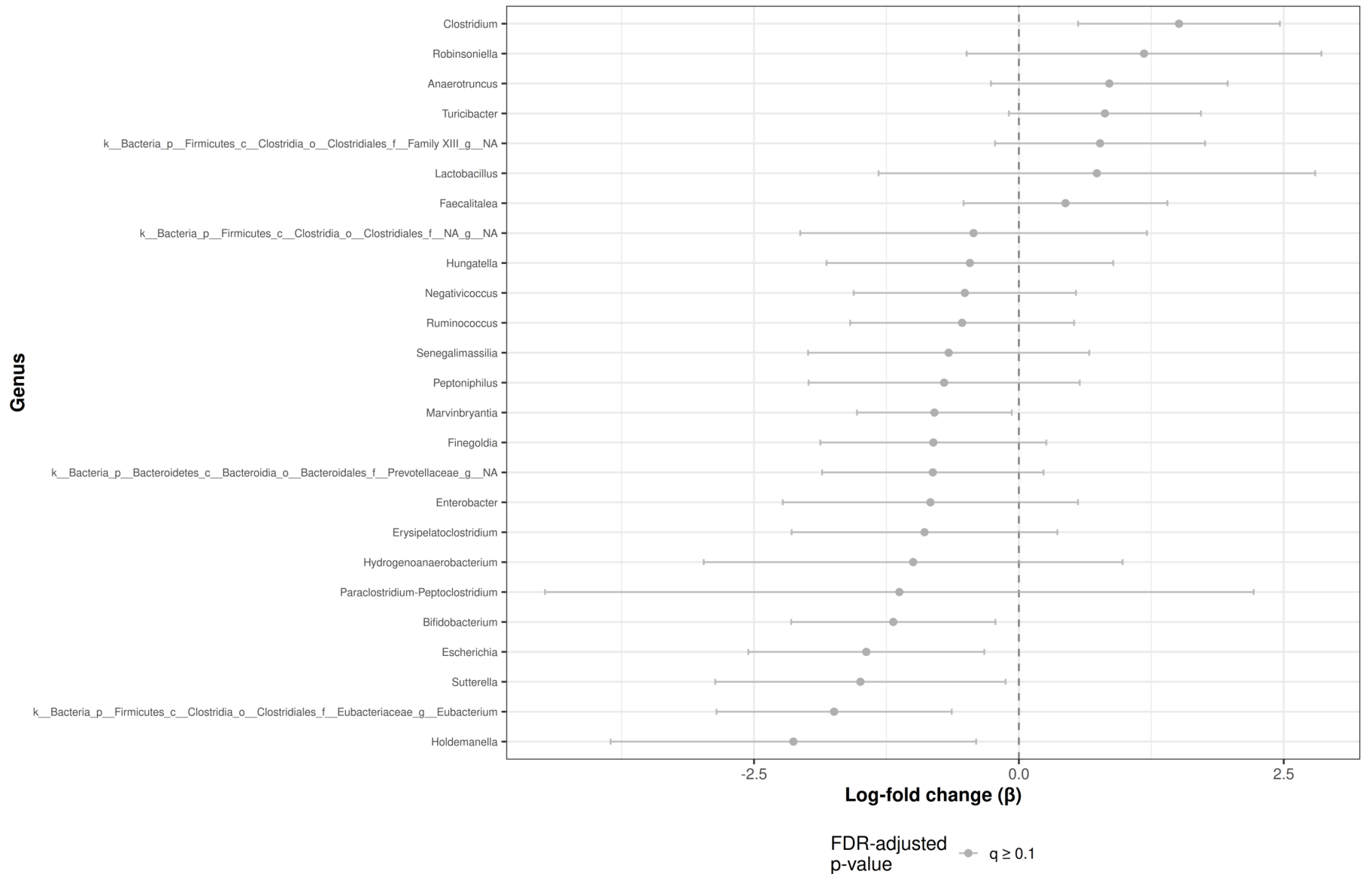
```
## MaAsLin2 butyrate responder: 1 significant associations (q < 0.1)
```

7.6.2 ANCOM-BC2 and MaAsLin2 Visualizations: Butyrate Responders vs Non-

Responders

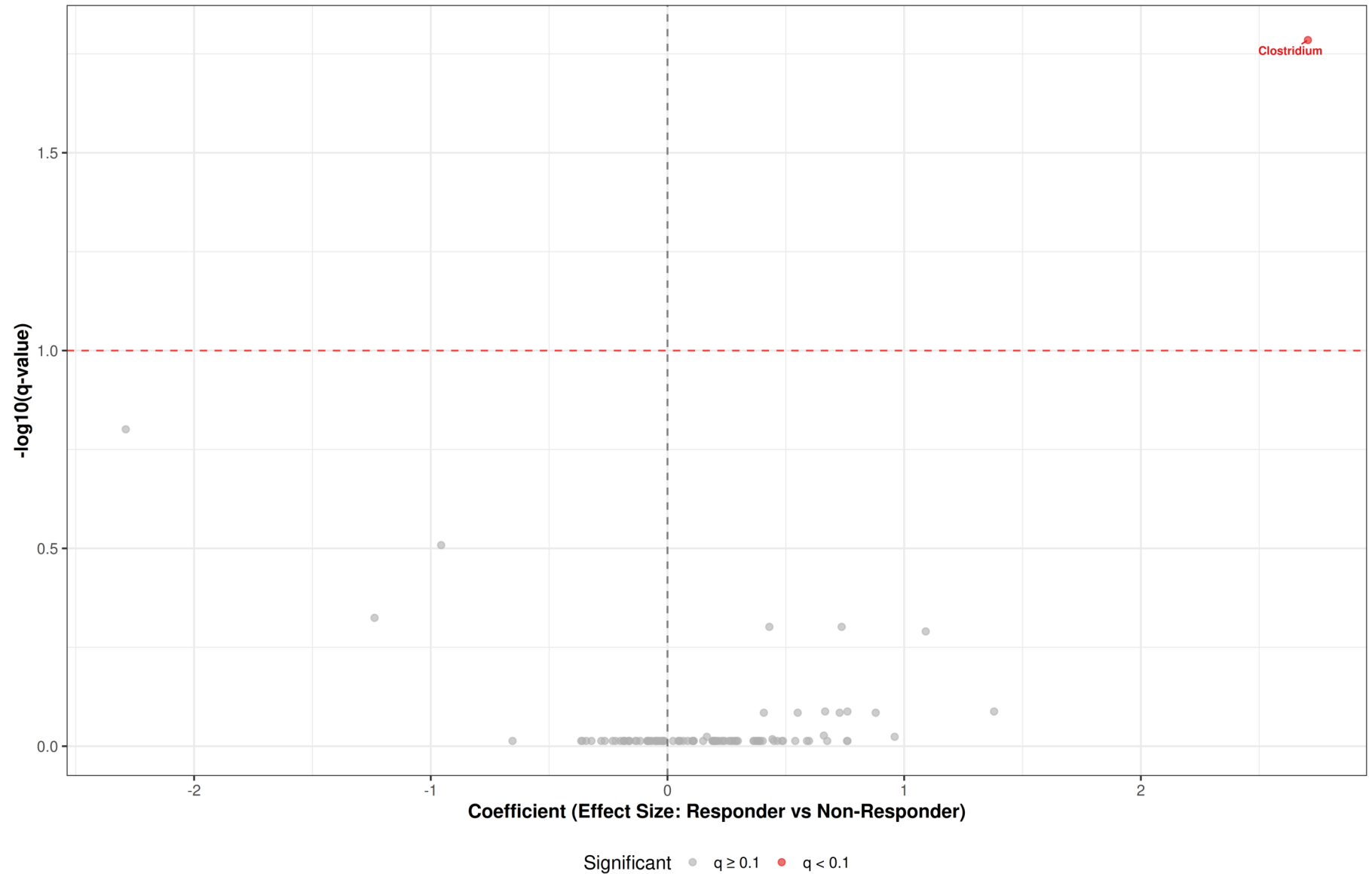
ANCOM-BC2: Butyrate Responders vs Non-Responders

Effect sizes (log-fold change: Responder vs Non-Responder) with 95% confidence intervals



MaAsLin2: Butyrate Responders vs Non-Responders

Effect Size vs Statistical Significance

Red dashed line: $q = 0.1$ threshold; Positive values indicate higher abundance in Responders

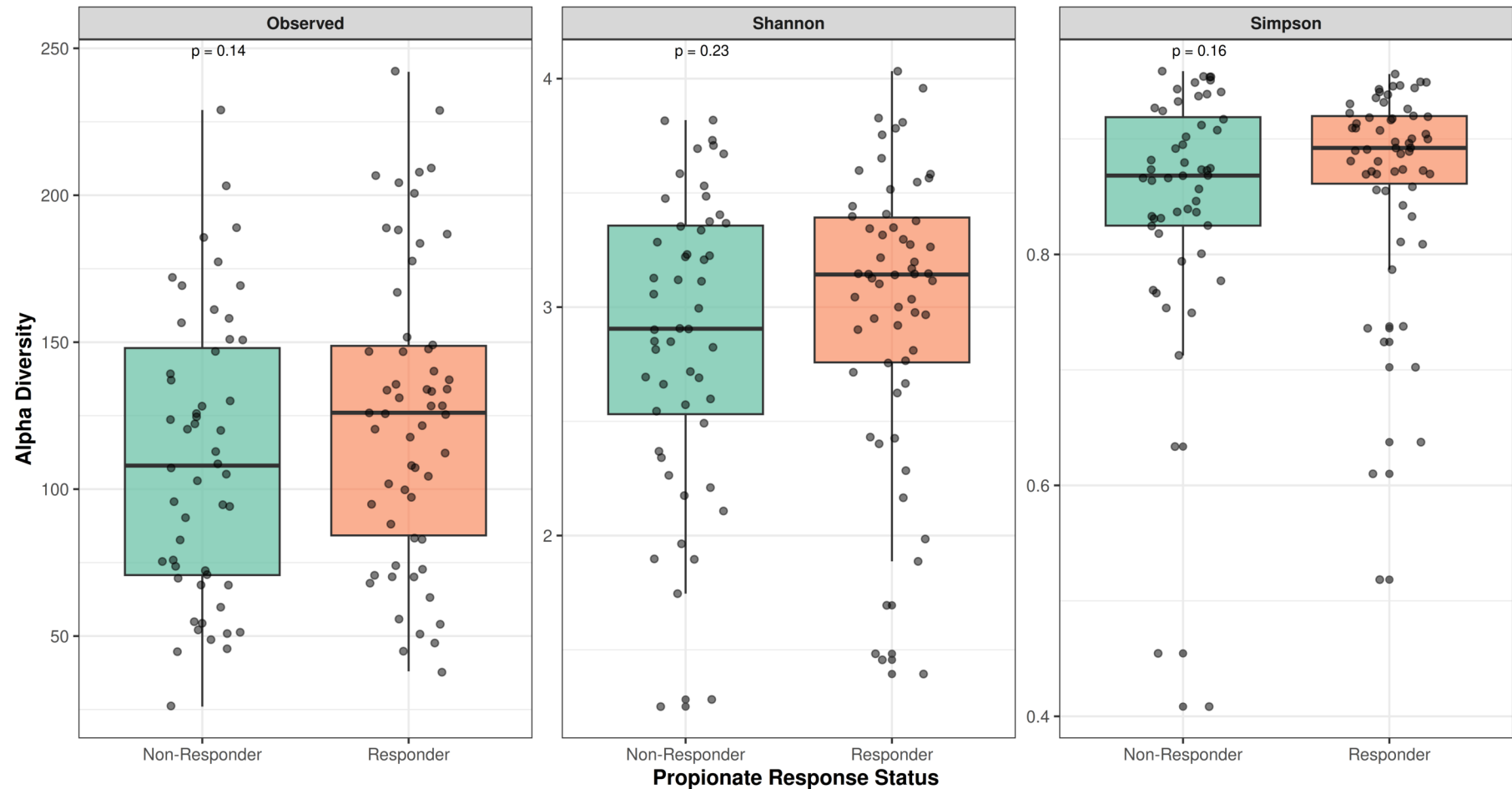
7.7 Microbial Composition Differences: Propionate Responders vs

Non-Responders

Following the same responder classification approach described in Section 7.6, we compare the microbial composition between propionate responders and non-responders at the 48H timepoint. Propionate responder status was determined using a median split of the delta change in propionate concentration (48H - 0H) for each carbohydrate condition separately. This analysis examines whether individuals with greater propionate production responses have distinct microbial communities compared to those with smaller responses.

Alpha Diversity: Propionate Responders vs Non-Responders (48H)

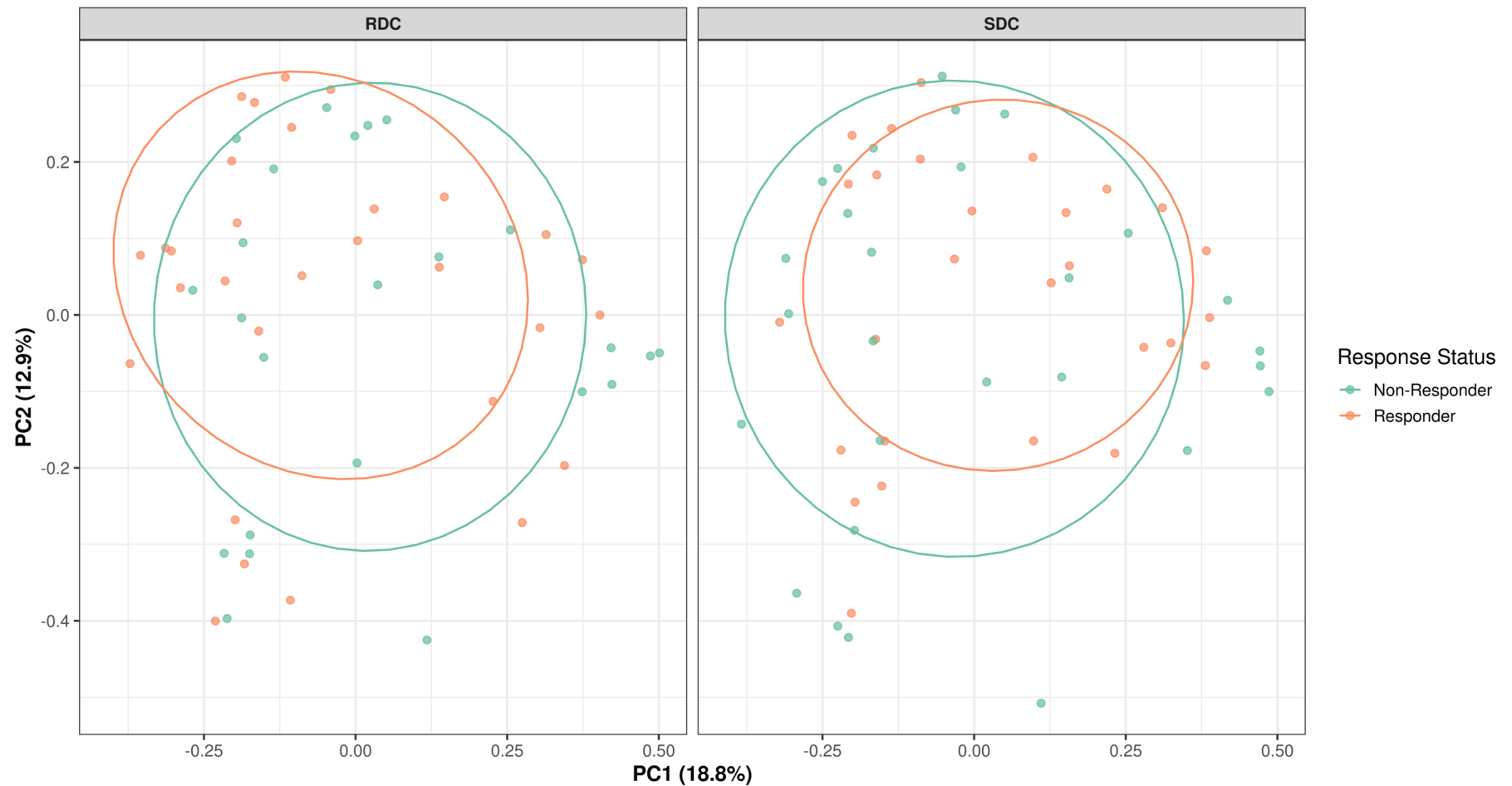
Comparison across RDC and SDC conditions (Wilcoxon test p-values shown)



```
##
## Wilcoxon Test Results: Propionate Responders vs Non-Responders (Alpha Diversity)
## =====
## # A tibble: 3 × 6
##   Metric   Group1      Group2   `p value` `p adjusted` Significance
##   <chr>    <chr>      <chr>    <dbl>     <dbl> <chr>
## 1 Observed Non-Responder Responder  0.141     0.225 ns
## 2 Shannon  Non-Responder Responder  0.225     0.225 ns
## 3 Simpson  Non-Responder Responder  0.155     0.225 ns
```

Beta Diversity: Propionate Responders vs Non-Responders (48H)

Bray-Curtis PCoA



```
##
## PERMANOVA: Propionate Responder Status (48H):
## Permutation test for adonis under reduced model
## Blocks: strata
## Permutation: free
## Number of permutations: 999
##
## vegan::adonis2(formula = dist_bray_prop_resp ~ responder_propionate * carbohydrate, data = meta_prop_resp, permutations =
999, strata = meta_prop_resp$subject_id)
##           Df SumOfSqs      R2      F Pr(>F)
## Model      3      1.695 0.04939 1.8359  0.041 *
## Residual 106     32.614 0.95061
## Total    109     34.308 1.00000
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

7.7.1 Differential Abundance Analysis: Propionate Responders vs Non-Responders

To identify specific taxa that differ between propionate responders and non-responders, we performed differential abundance analysis using ANCOM-BC2 and MaAsLin2.

```
## ANCOM-BC2 propionate responder: 0 significant taxa (q < 0.1)
```

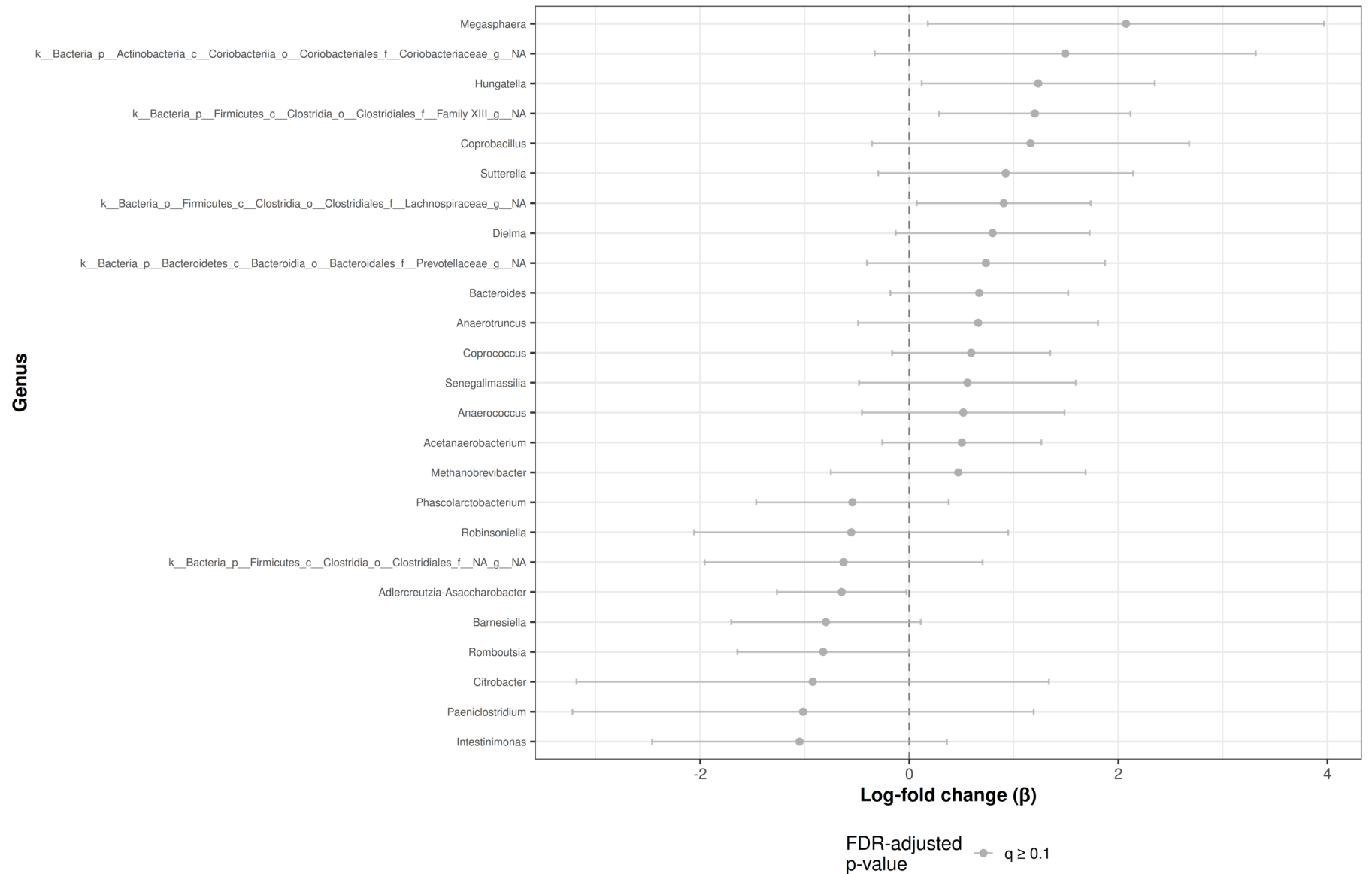
```
## MaAsLin2 propionate responder: 1 significant associations (q < 0.1)
```

7.7.2 ANCOM-BC2 and MaAsLin2 Visualizations: Propionate Responders vs Non-

Responders

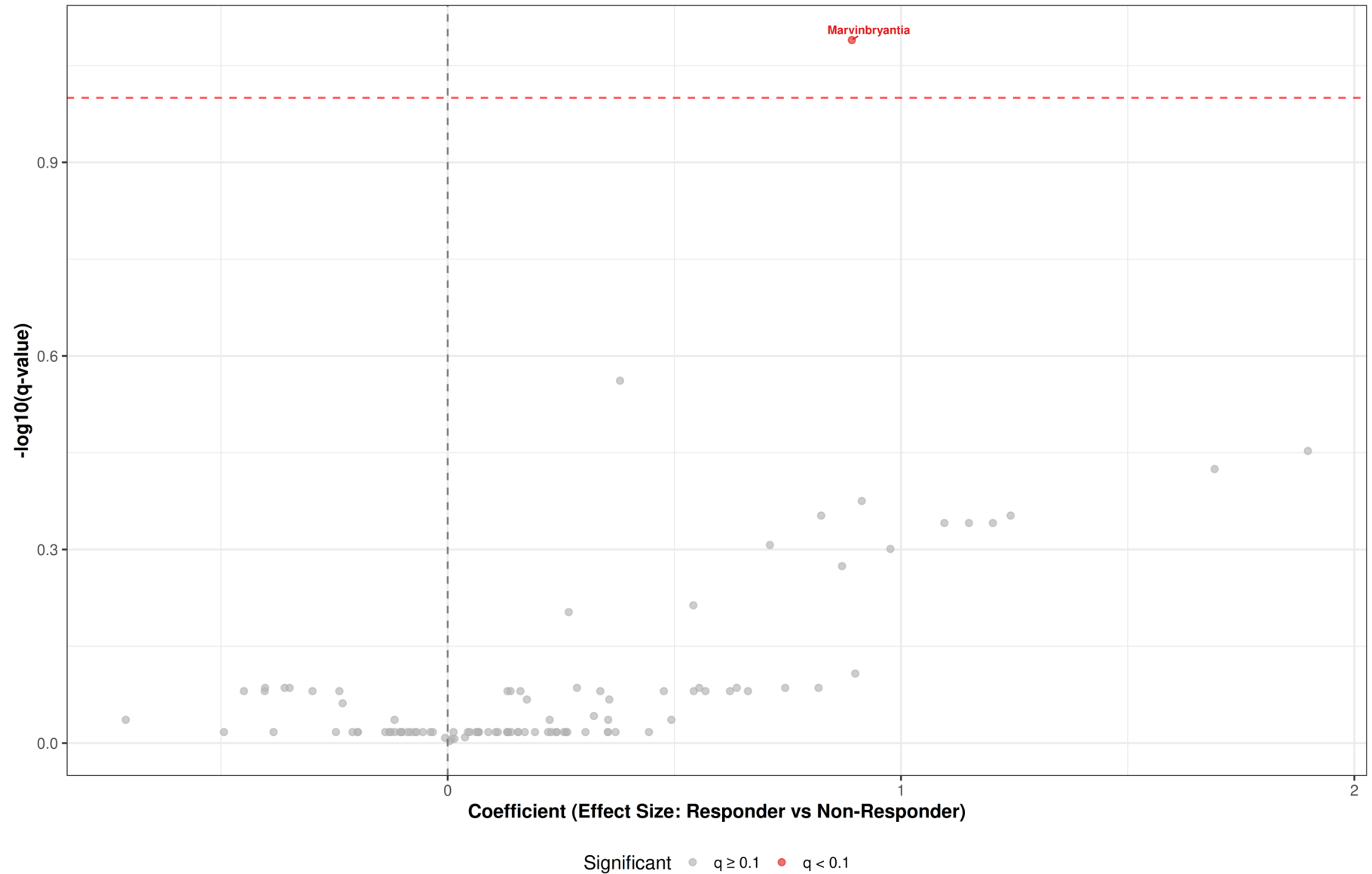
ANCOM-BC2: Propionate Responders vs Non-Responders

Effect sizes (log-fold change: Responder vs Non-Responder) with 95% confidence intervals



MaAsLin2: Propionate Responders vs Non-Responders

Effect Size vs Statistical Significance

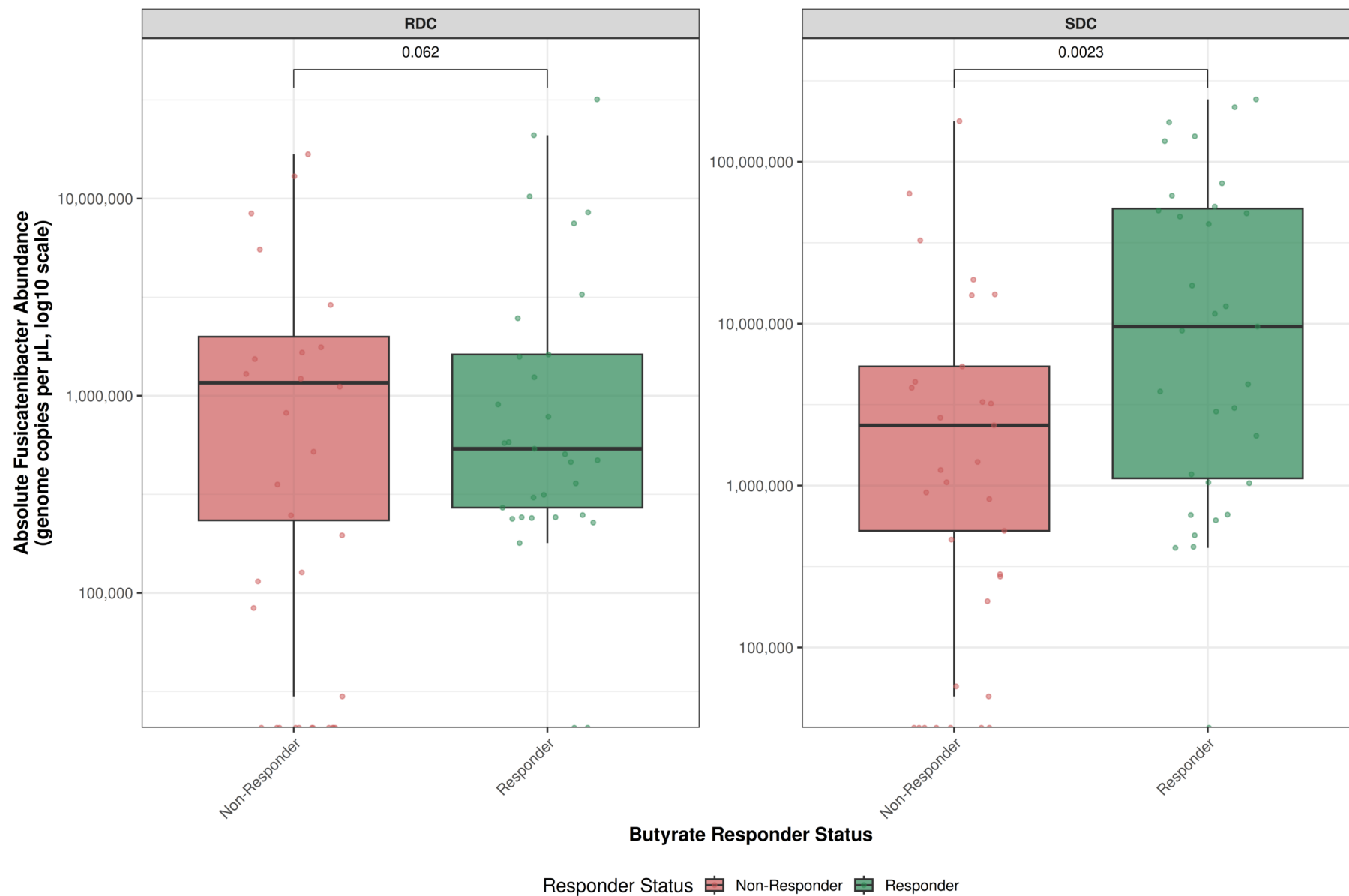


Red dashed line: q = 0.1 threshold; Positive values indicate higher abundance in Responders

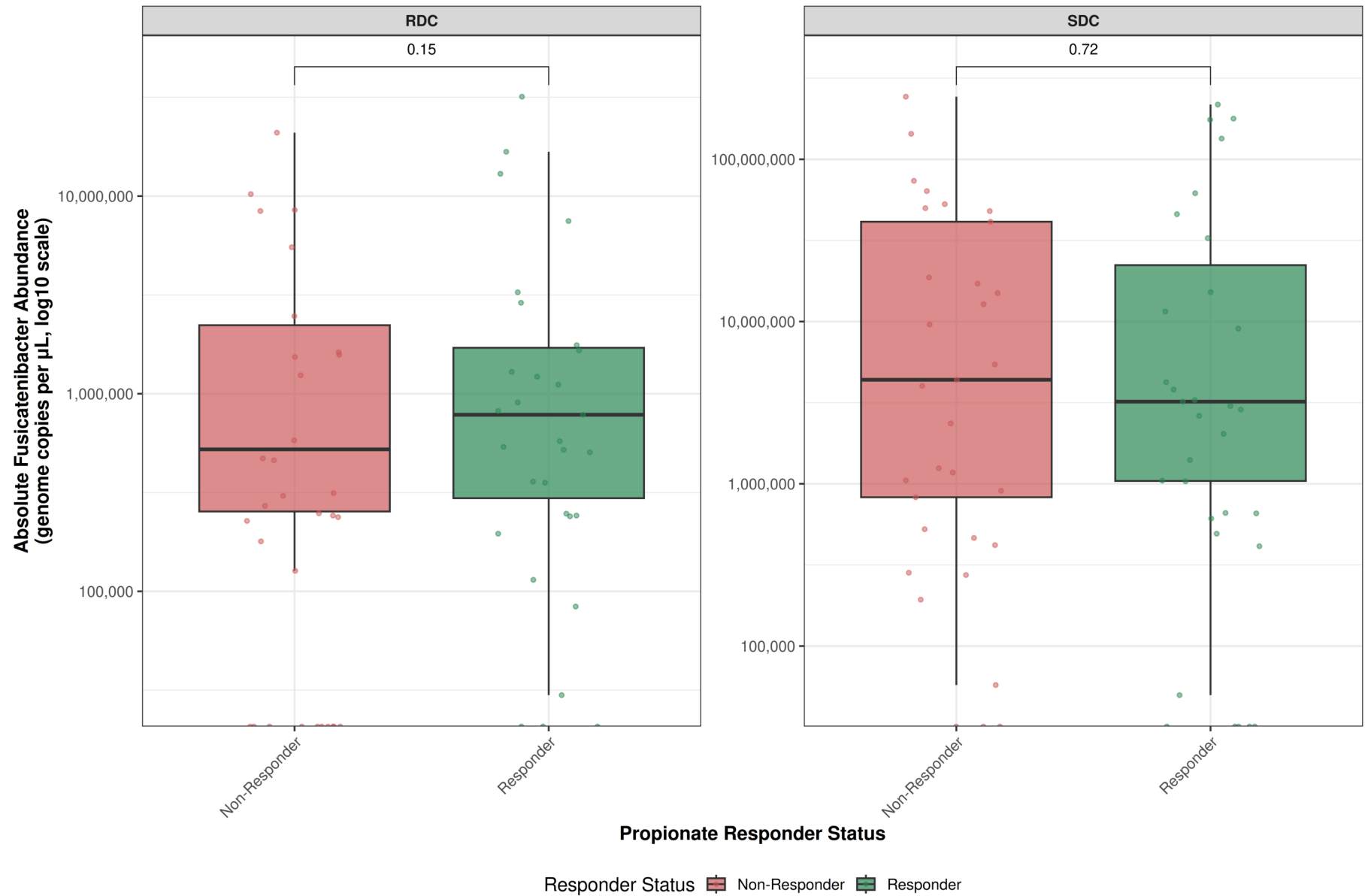
7.8 Absolute Fusicatenibacter Abundance: Responders vs Non-

Responders

To determine whether absolute *Fusicatenibacter* abundance differs between SCFA responders and non-responders, we compared absolute abundance levels (genome copies per μL) at the 48H timepoint. This analysis uses qPCR-derived total bacterial load data integrated with relative abundance measurements to calculate absolute *Fusicatenibacter* abundance.

Absolute Fusicatenibacter Abundance: Butyrate Responders vs Non-Responders48H timepoint, genome copies per μL (log10 scale)

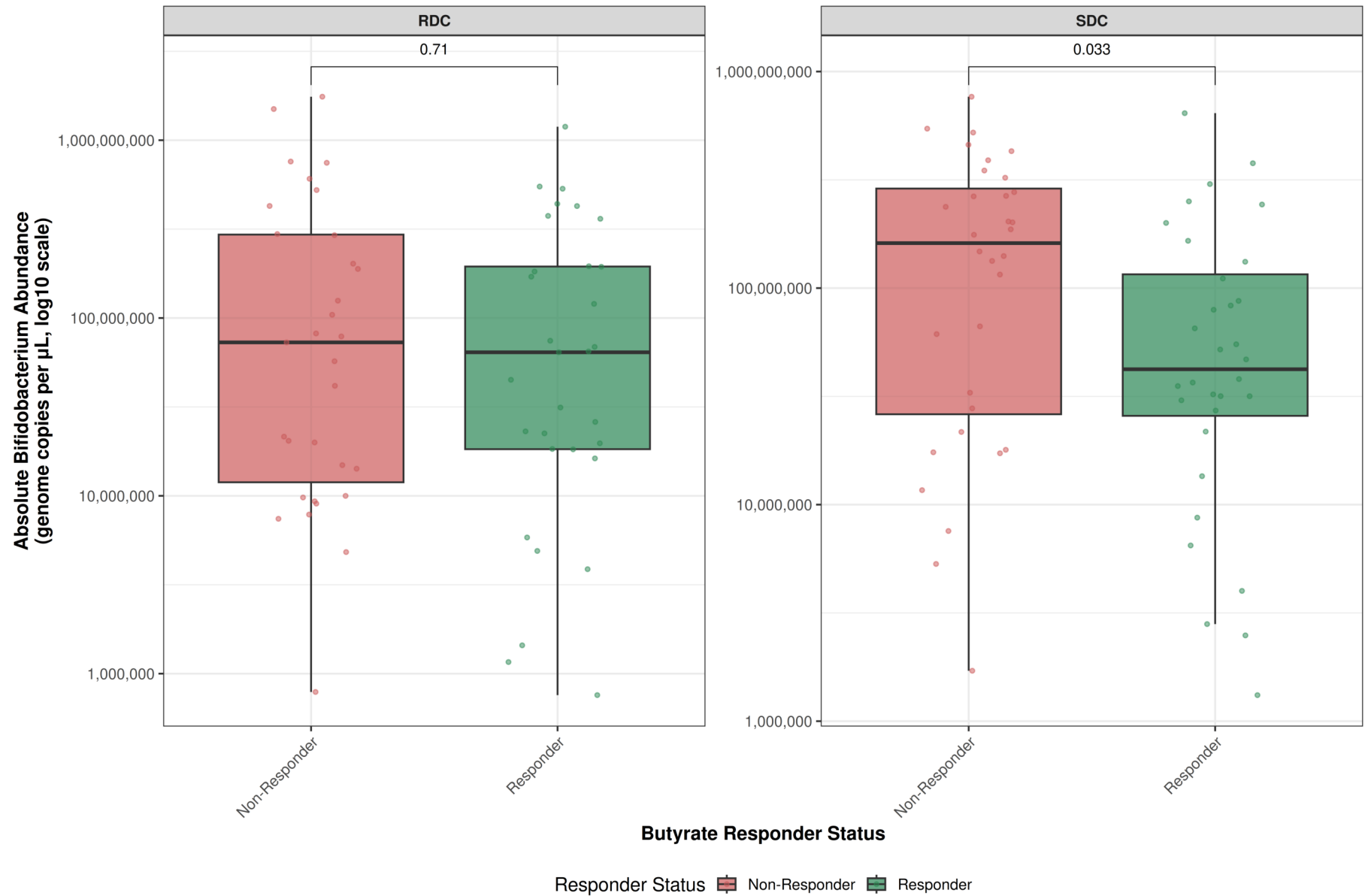

```
##
## ### Butyrate Responder Analysis:
## # A tibble: 4 × 6
##   responder_butyrate carbohydrate median_abs mean_abs se_abs n
##   <chr>                <chr>          <dbl>    <dbl>    <dbl> <int>
## 1 Non-Responder       RDC            195794.  1858093.  707443.  31
## 2 Non-Responder       SDC            977671.  11121706.  5837706.  32
## 3 Responder          RDC            505336.  3123972.  1231200.  31
## 4 Responder          SDC            9329027.  42798736.  11817629.  32
##
## Wilcoxon Test Results:
## # A tibble: 1 × 7
##   .y.                group1      group2      n1    n2 statistic      p
## * <chr>            <chr>      <chr>    <int> <int>    <dbl>    <dbl>
## 1 fusicatenibacter_abs_abund Non-Responder Responder    63    63    1323 0.00123
```

Absolute Fusicatenibacter Abundance: Propionate Responders vs Non-Responders48H timepoint, genome copies per μL (log10 scale)

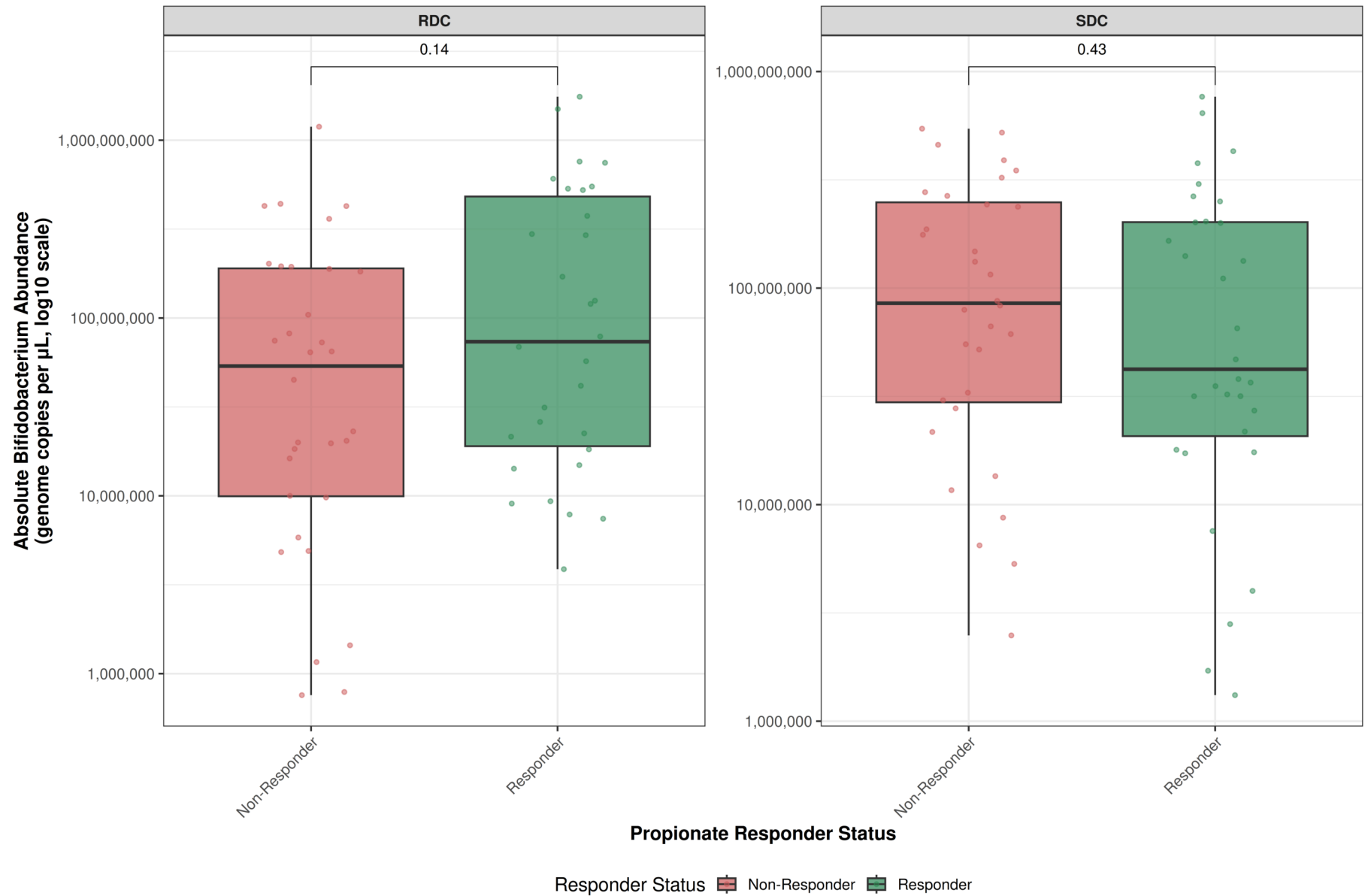
```
##
## ### Propionate Responder Analysis:
## # A tibble: 4 × 6
##   responder_propionate carbohydrate median_abs mean_abs se_abs n
##   <chr>                <chr>          <dbl>    <dbl>    <dbl> <int>
## 1 Non-Responder       RDC            259365.  2053187.  776525.  32
## 2 Non-Responder       SDC            3185150. 25402241.  8924278.  32
## 3 Responder           RDC            556837.  2958068.  1217298.  30
## 4 Responder           SDC            2746683. 28518201. 10493819.  32
##
## Wilcoxon Test Results:
## # A tibble: 1 × 7
##   .y.                group1      group2      n1      n2 statistic      p
## * <chr>            <chr>      <chr>    <int> <int>    <dbl> <dbl>
## 1 fusicatenibacter_abs_abund Non-Responder Responder    64    62    1853 0.523
```

7.9 Absolute Bifidobacterium Abundance: Responders vs Non-Responders

To determine whether absolute *Bifidobacterium* abundance differs between SCFA responders and non-responders, we compared absolute abundance levels (genome copies per μL) at the 48H timepoint. This analysis uses qPCR-derived total bacterial load data integrated with relative abundance measurements to calculate absolute *Bifidobacterium* abundance.

Absolute Bifidobacterium Abundance: Butyrate Responders vs Non-Responders48H timepoint, genome copies per μL (log10 scale)

```
##
## ### Butyrate Responder Analysis:
## # A tibble: 4 × 6
##   responder_butyrate carbohydrate median_abs   mean_abs   se_abs     n
##   <chr>                <chr>          <dbl>     <dbl>     <dbl> <int>
## 1 Non-Responder       RDC           72946996. 258322171. 77015069.    31
## 2 Non-Responder       SDC          161960340. 200714144. 33989331.    32
## 3 Responder           RDC           64155493. 169288167. 45645934.    31
## 4 Responder           SDC           42425998. 100571916. 24285853.    32
##
## Wilcoxon Test Results:
## # A tibble: 1 × 7
##   .y.                group1      group2     n1    n2 statistic     p
## * <chr>            <chr>      <chr>    <int> <int>     <dbl> <dbl>
## 1 bifidobacterium_abs_abund Non-Responder Responder    63    63      2319 0.103
```

Absolute Bifidobacterium Abundance: Propionate Responders vs Non-Responders48H timepoint, genome copies per μL (log10 scale)

```
##
## ### Propionate Responder Analysis:
## # A tibble: 4 × 6
##   responder_propionate carbohydrate median_abs mean_abs se_abs n
##   <chr>                <chr>          <dbl>    <dbl>   <dbl> <int>
## 1 Non-Responder        RDC          54543035. 139742370. 41478245. 32
## 2 Non-Responder        SDC          85154019. 156785330. 28269346. 32
## 3 Responder            RDC          73777515. 292805487. 79562440. 30
## 4 Responder            SDC          42425998. 144500730. 33245101. 32
##
## Wilcoxon Test Results:
## # A tibble: 1 × 7
##   .y.          group1      group2    n1    n2 statistic    p
## * <chr>      <chr>      <chr>   <int> <int>   <dbl> <dbl>
## 1 bifidobacterium_abs_abund Non-Responder Responder    64    62    1855 0.531
```

8. Summary

This integrated analysis brings together SCFA metabolomics and 16S rRNA gene microbiome data to provide a comprehensive view of how carbohydrate interventions affect microbial metabolism and community structure in adolescents with and without obesity. Below we summarize the key findings from each major analysis component.

8.1 SCFA Metabolomics Findings

Carbohydrate Type Effects: - Both rapid digestible carbohydrate (RDC) and slow digestible carbohydrate (SDC) significantly increased SCFA production (butyrate, propionate, acetate) over 48 hours compared to no-carbohydrate controls. - Delta changes (48H - 0H) showed significant differences between carbohydrate conditions across multiple SCFAs, with both RDC and SDC stimulating substantial SCFA production. - The magnitude and timing of SCFA responses varied by carbohydrate type, as evidenced by significant carbohydrate × timepoint interactions in mixed-effects models.

Group Differences: - Limited significant differences were observed between Control (Healthy) and Case (Obese) groups in SCFA production capacity, suggesting that obesity status alone does not dramatically alter the fundamental ability of gut microbiota to ferment these carbohydrate substrates. - Mixed-effects models showed minimal significant main effects of group or major interactions involving group, indicating similar fermentation capacity between groups.

Interpersonal Variation: - Pronounced subject-to-subject variability in SCFA production was observed, with some individuals consistently showing high or low SCFA responses across carbohydrate conditions. - This variability was evident even after accounting for group and carbohydrate type effects, highlighting the need for personalized dietary approaches.

8.2 Microbiome Composition Findings

Alpha Diversity: - Alpha diversity metrics (Shannon, Simpson, Observed richness) showed significant differences between Healthy and Obese groups at the 48H timepoint, particularly in the no-carbohydrate control condition. - At baseline (0H), no significant differences in alpha diversity were observed between groups across carbohydrate conditions. - By 48H, Healthy individuals showed higher alpha diversity than Obese individuals in the no-carbohydrate condition, suggesting that carbohydrate challenge may reveal group differences in microbial diversity.

Beta Diversity and Community Structure: - PERMANOVA analyses demonstrated significant effects of group, carbohydrate type, and timepoint on microbial community composition (Bray-Curtis distance). - Community structure differences were most pronounced at 48H, with significant effects of both group and carbohydrate type on beta diversity. - The interaction between group and carbohydrate type was significant, indicating that the response to different carbohydrates varies by obesity status.

Differential Abundance: - ANCOM-BC2 and MaAsLin2 analyses identified multiple taxa that differed significantly between carbohydrate conditions, groups, and timepoints. - At 48H, significant differences were observed between RDC and SDC conditions, with distinct taxa associated with each carbohydrate type. - These findings suggest that carbohydrate structure and digestibility shape microbial community composition in predictable ways.

8.3 Taxon–SCFA Relationships

Butyrate Associations: - ANCOM-BC2 identified 133 taxa significantly associated with butyrate concentrations ($q < 0.1$) after adjusting for carbohydrate effects. - MaAsLin2 identified 400 significant associations between taxa and butyrate concentrations ($q < 0.1$).

Propionate Associations: - ANCOM-BC2 identified 104 taxa significantly associated with propionate concentrations ($q < 0.1$) after adjusting for carbohydrate effects. - MaAsLin2 identified 397 significant associations between taxa and propionate concentrations ($q < 0.1$).

Fusicatenibacter–SCFA Correlations: - *Fusicatenibacter* relative abundance showed a significant correlation with butyrate concentration at 48H in the SDC condition ($R^2 = 0.037$, $p = 0.1569$). - *Fusicatenibacter* relative abundance also correlated with propionate concentration at 48H in the SDC condition ($R^2 = 0.004$, $p = 0.6546$). - These correlations were specific to the 48H timepoint and SDC carbohydrate condition, suggesting that *Fusicatenibacter* may play a role in SCFA production in response to slow-digestible carbohydrates.

8.4 Responder vs Non-Responder Analysis

Responder Classification: - Responder status was defined separately for butyrate and propionate based on median delta SCFA changes (48H - 0H) within each carbohydrate condition. - This data-driven approach identified individuals who showed substantial SCFA responses relative to their peers in the same condition.

Overlap in Responder Status: - Of 48 subjects with complete data, 12 (25.0%) were responders for both butyrate and propionate. - 12 (25.0%) were non-responders for both SCFAs. - This partial overlap suggests that butyrate and propionate responses are partially independent, with some individuals showing selective responses to one SCFA but not the other.

Microbial Composition Differences: - Alpha diversity comparisons between responder and non-responder groups at 48H revealed significant differences for both butyrate and propionate responders. - Differential abundance analyses (ANCOM-BC2 and MaAsLin2) identified distinct taxa associated with responder vs non-responder status for both butyrate and propionate. - These findings indicate that microbial community composition is predictive of SCFA response capacity, with specific taxa enriched in responder groups.

8.5 Integrated Conclusions

Key Insights: 1. **Carbohydrate structure matters:** Both RDC and SDC stimulate SCFA production, but the specific taxa and metabolic pathways involved differ, as evidenced by distinct differential abundance patterns and community structure changes.

2. **Obesity status has limited impact on fermentation capacity:** While some differences in baseline microbial diversity were observed, the fundamental ability to ferment carbohydrates and produce SCFAs appears similar between Healthy and Obese groups, suggesting that obesity-related microbiome changes may not directly impair carbohydrate fermentation.
3. **Individual variation is substantial:** The pronounced interpersonal variation in SCFA responses, combined with distinct microbial signatures in responder vs non-responder groups, highlights the importance of personalized nutrition approaches.
4. **Taxon–SCFA relationships are context-dependent:** Associations between specific taxa (e.g., *Fusicatenibacter*) and SCFA concentrations are strongest at specific timepoints (48H) and carbohydrate conditions (SDC), emphasizing the importance of considering temporal and dietary context in microbiome–metabolite relationships.
5. **Responder phenotypes have distinct microbial signatures:** The identification of microbial taxa associated with responder status suggests that microbiome composition may be predictive of individual SCFA response capacity, opening possibilities for microbiome-based precision nutrition strategies.

Clinical and Translational Implications: - The identification of responder vs non-responder groups, combined with distinct microbial signatures, suggests that microbiome profiling could inform personalized dietary recommendations. - The finding that *Fusicatenibacter* correlates with SCFA production in the SDC condition suggests that this taxon may be a key player in slow-digestible carbohydrate metabolism. - The partial independence of butyrate and propionate responder status indicates that different microbial pathways may drive production of these SCFAs, with implications for targeted dietary interventions.

Future Directions: - Functional metagenomics and metabolomics could identify the specific genes and pathways that drive SCFA production in responder groups. - Longitudinal studies could determine whether responder status is stable over time and whether dietary interventions can shift individuals from non-responder to responder status. - Clinical trials could test whether microbiome-based stratification improves the efficacy of dietary interventions for obesity management.